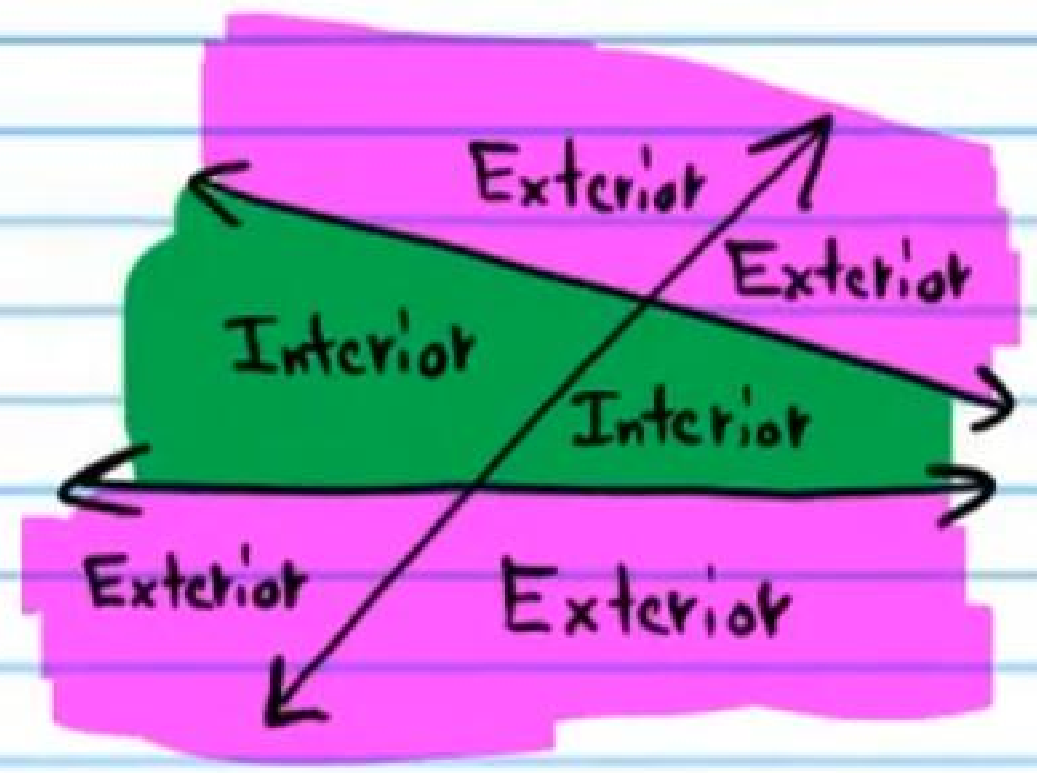
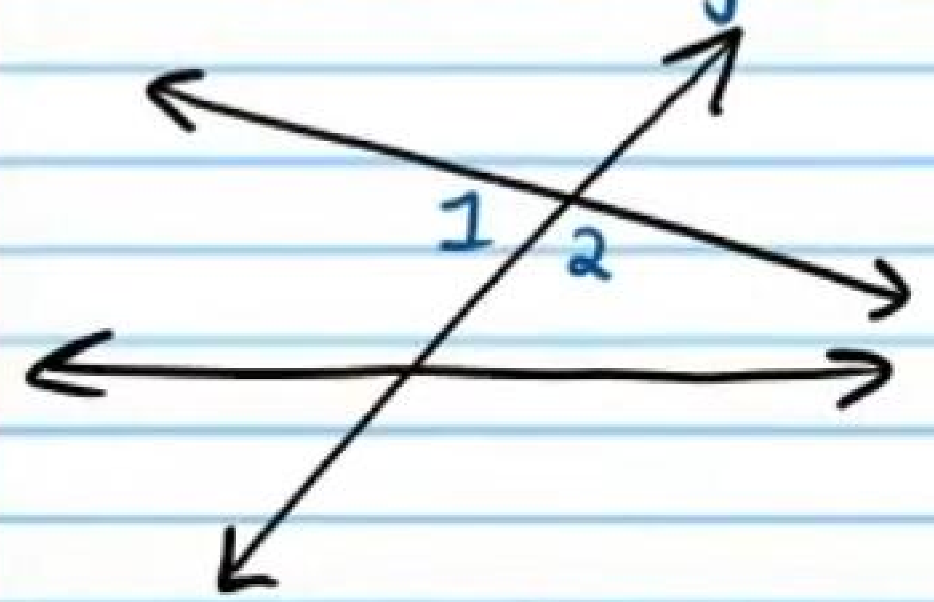
Name
Date
Course

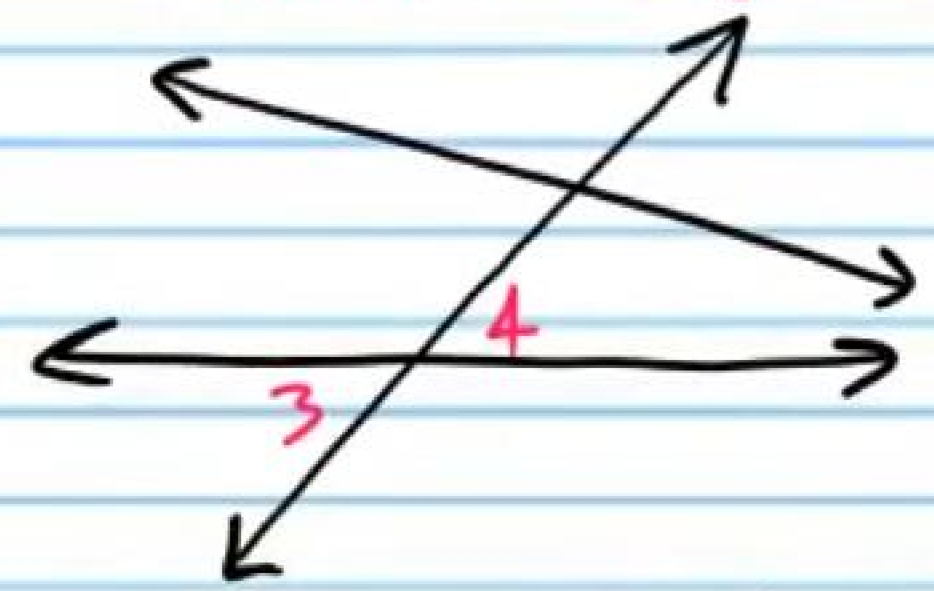
Angles Formed by Transversals



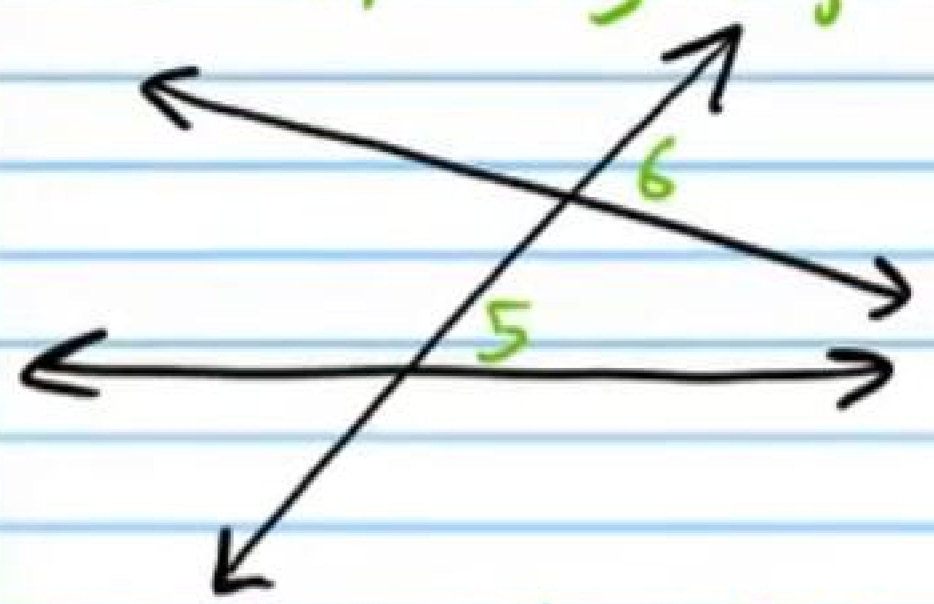
Adjacent Angles



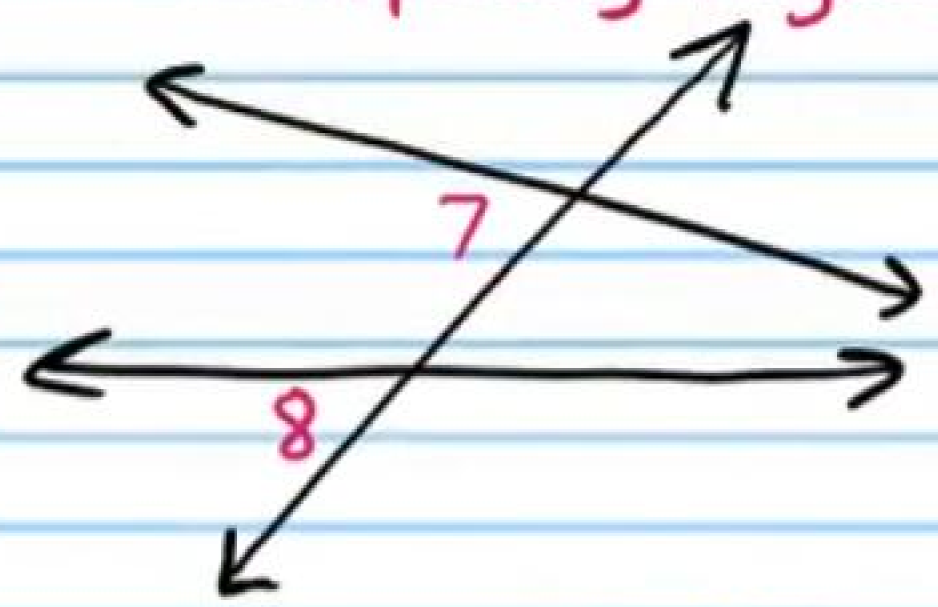
Vertical Angles



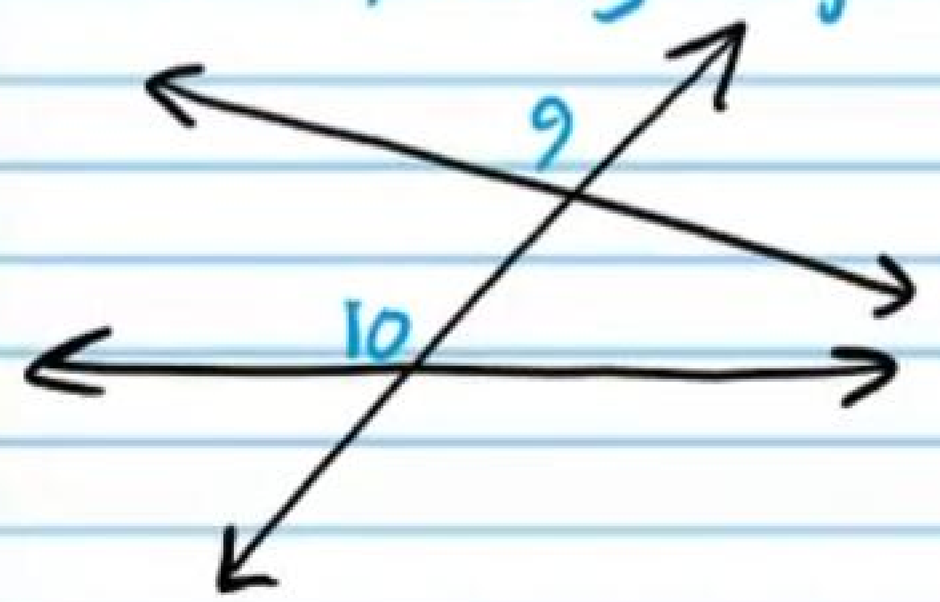
Corresponding Angles



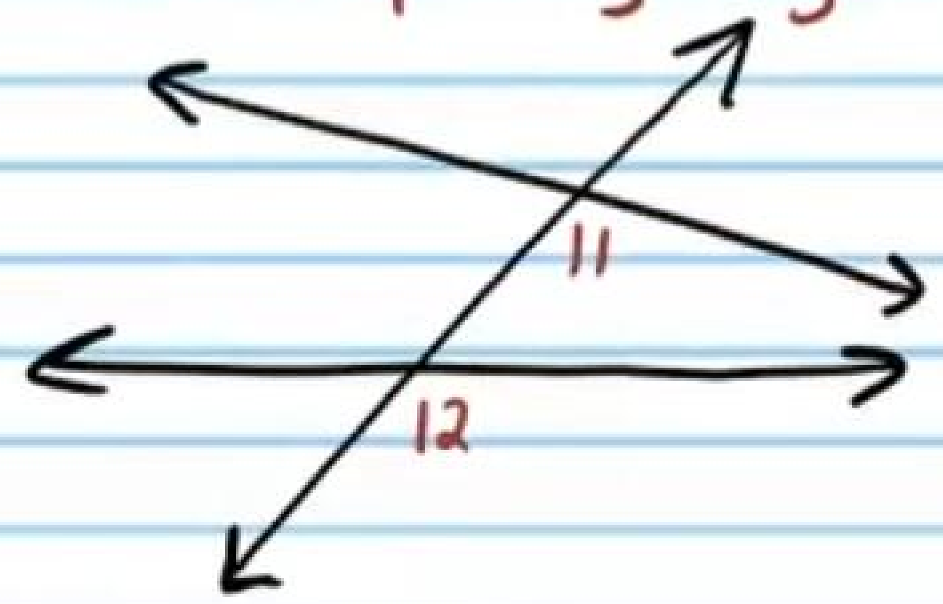
Corresponding Angles



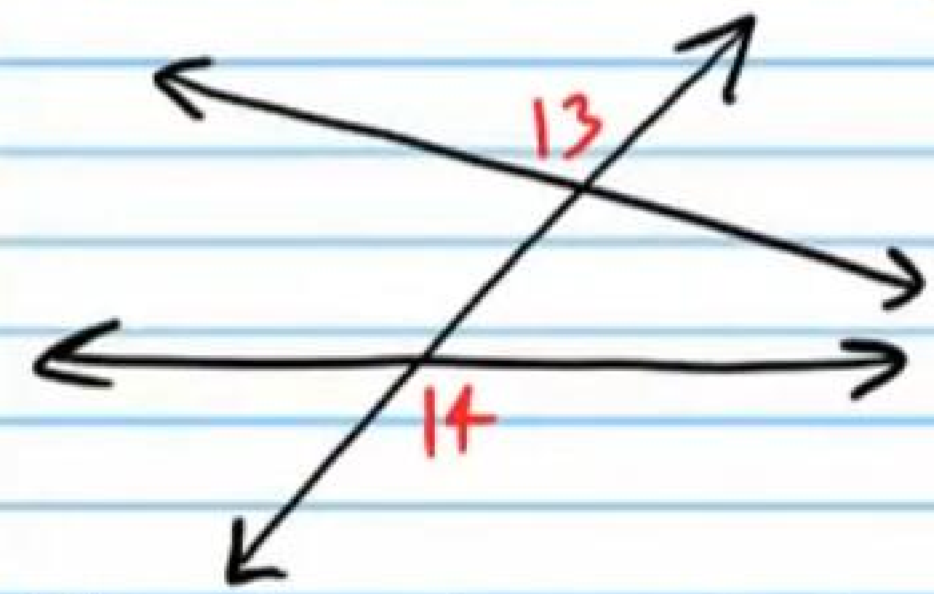
Corresponding Angles



Corresponding Angles



Alternate Exterior Angles



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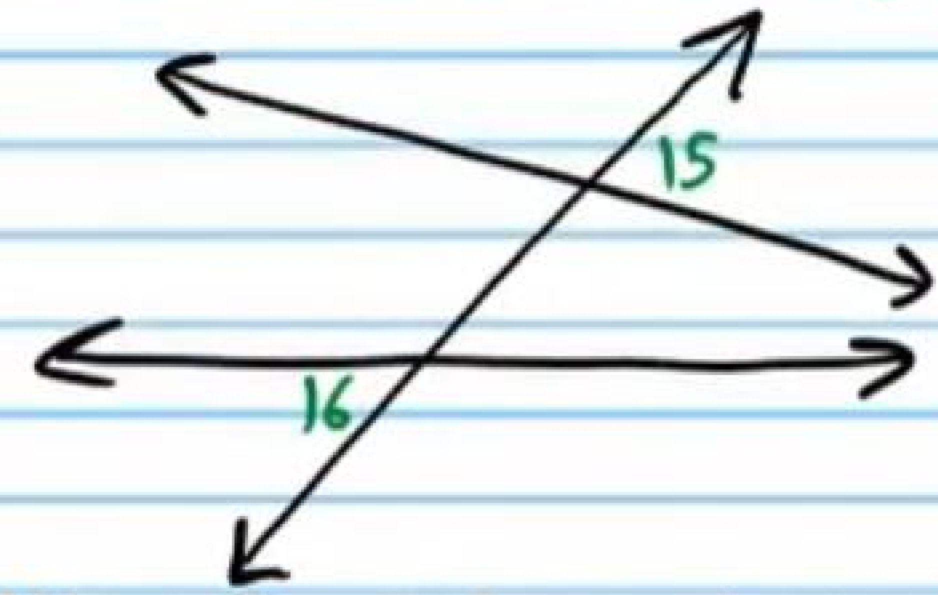
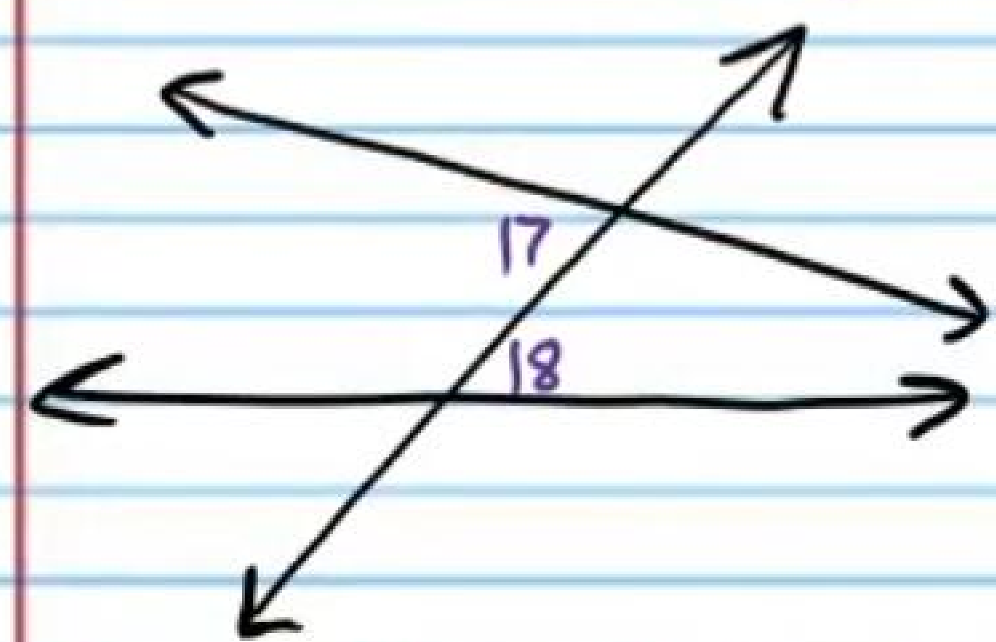
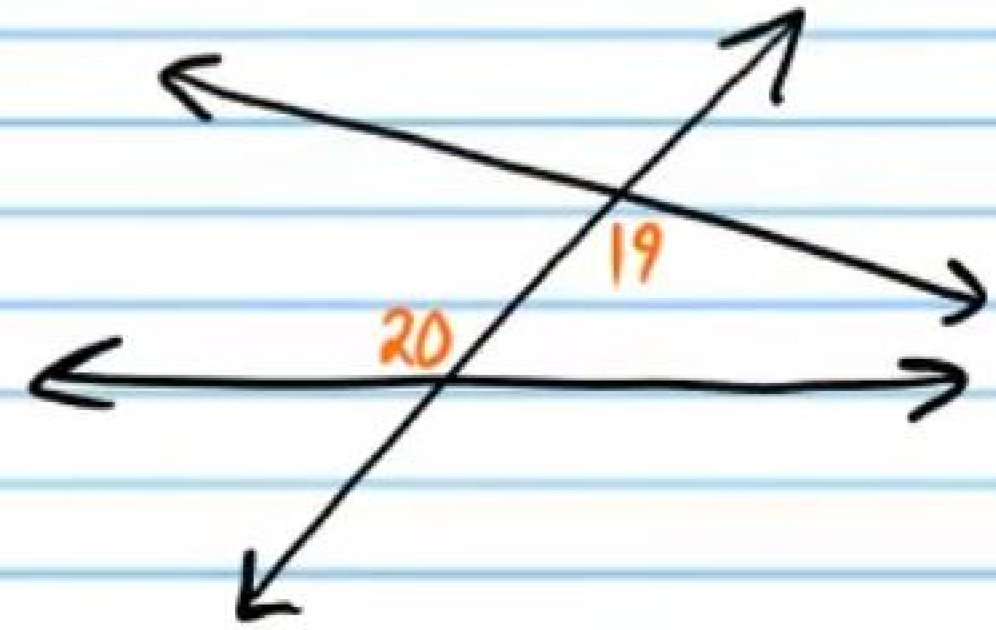


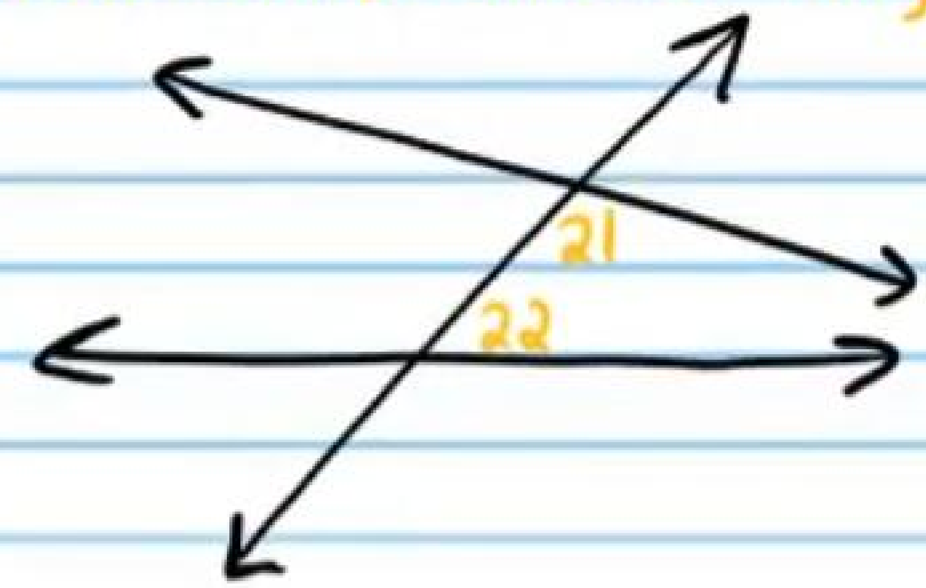
Alternate Interior Angles



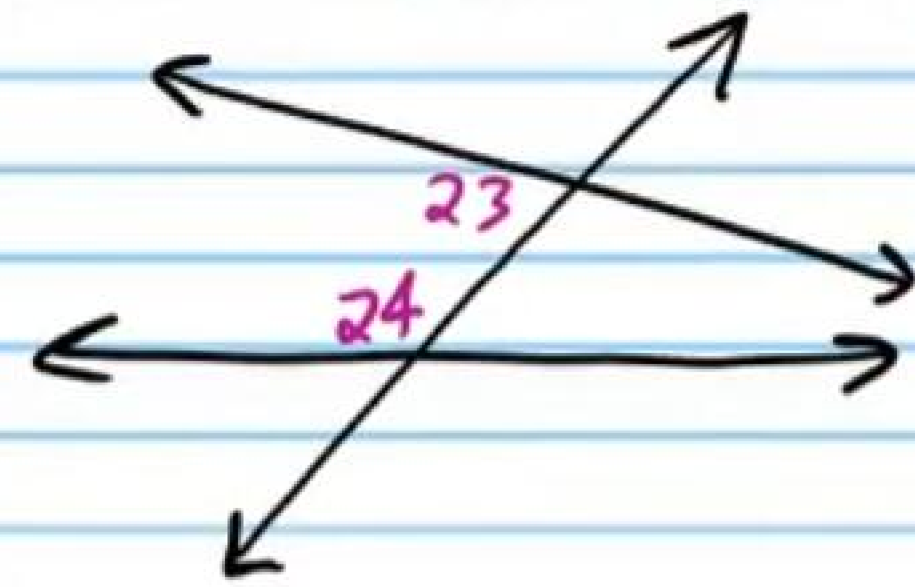
Alternate Interior Angles



Consecutive Interior Angles

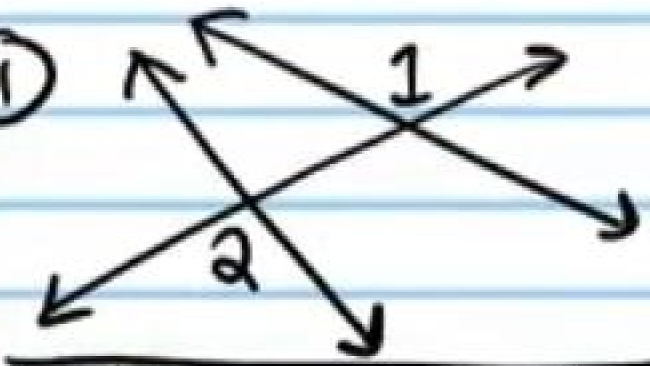


Consecutive Interior Angles

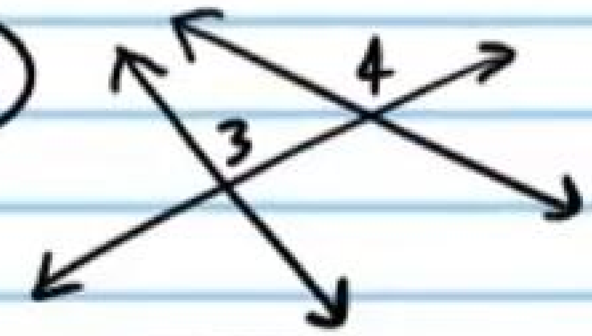


Practice

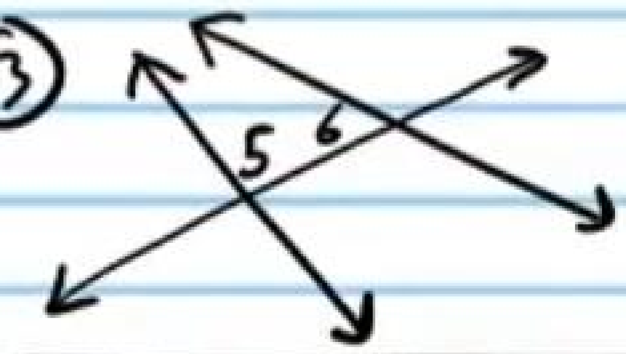
Name the angle pairs.

① 

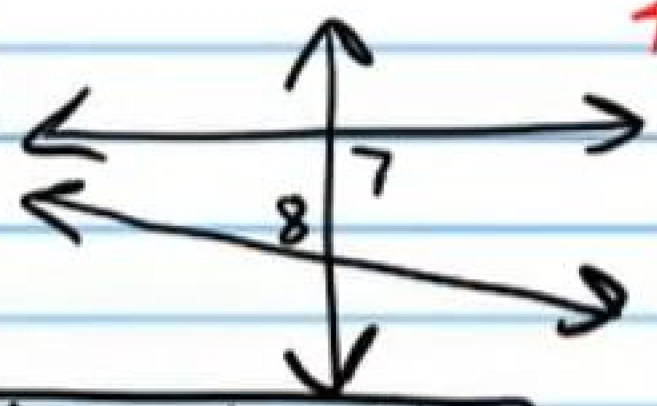
Alt. Ext. ∠'s

② 

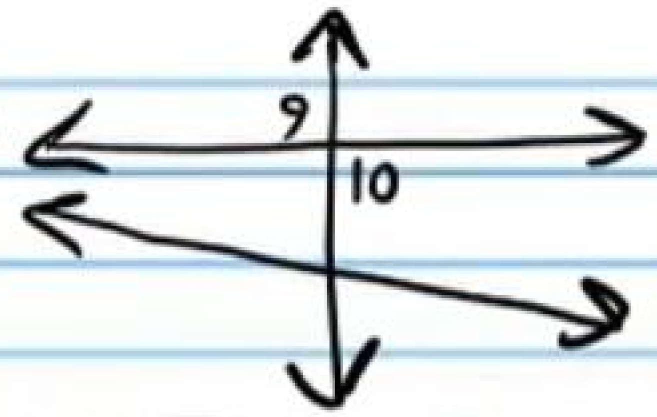
Corr. ∠'s

③ 

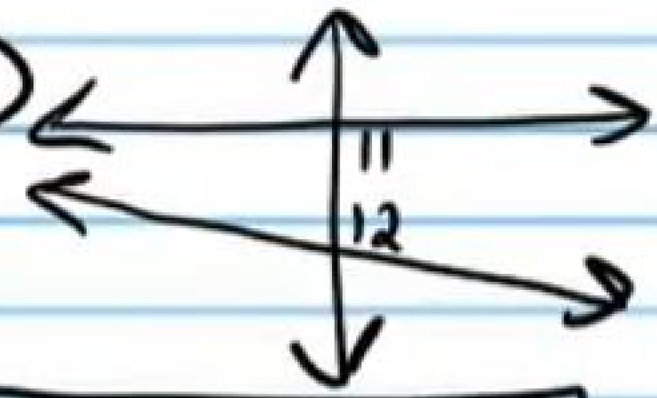
Cons. Int. ∠'s

④


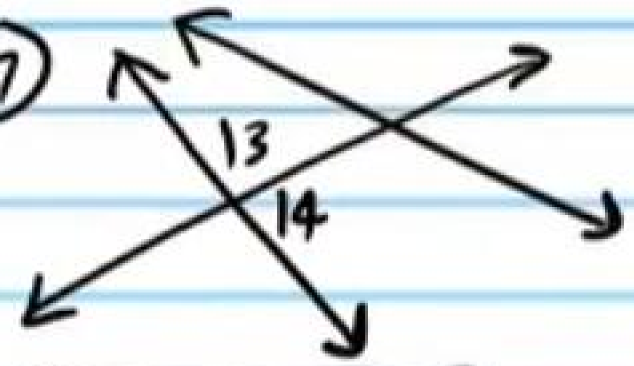
$\boxed{\text{Alt. Int. } \angle\text{'s}}$

⑤


$\boxed{\text{Vert. } \angle\text{'s}}$

⑥


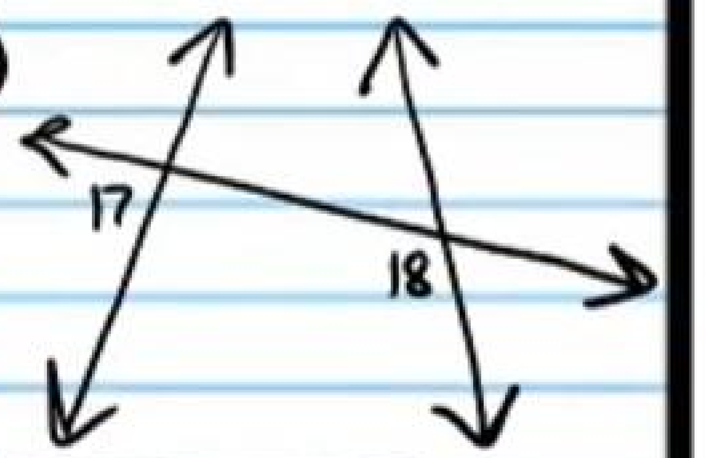
$\boxed{\text{Cons. Int } \angle\text{'s}}$

⑦


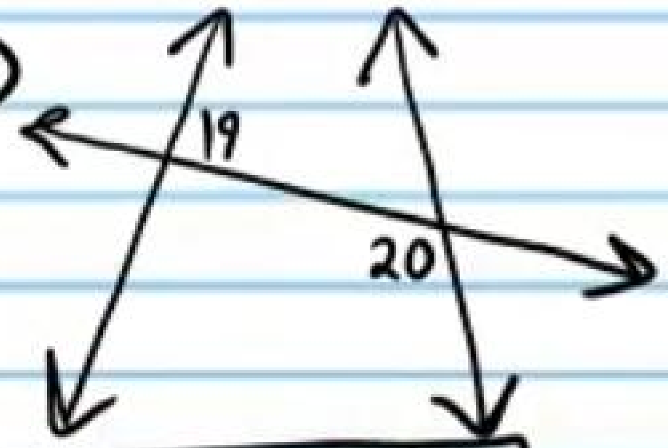
$\boxed{\text{Adj } \angle\text{'s}}$

⑧

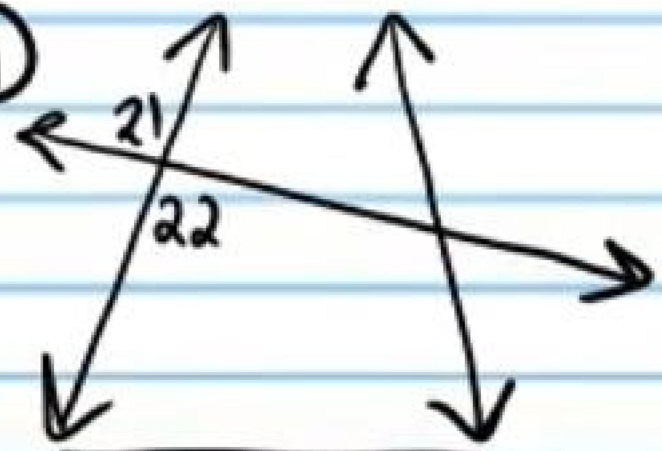

$\boxed{\text{Cons. Int. } \angle\text{'s}}$

⑨


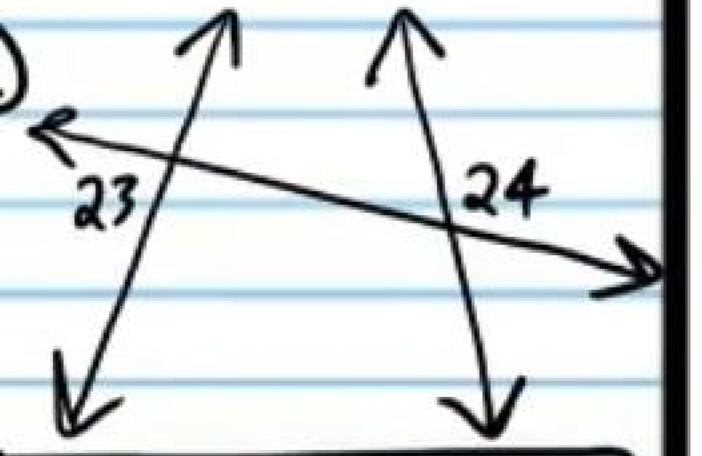
$\boxed{\text{Corr. } \angle\text{'s}}$

⑩


$\boxed{\text{Alt. Int. } \angle\text{'s}}$

⑪


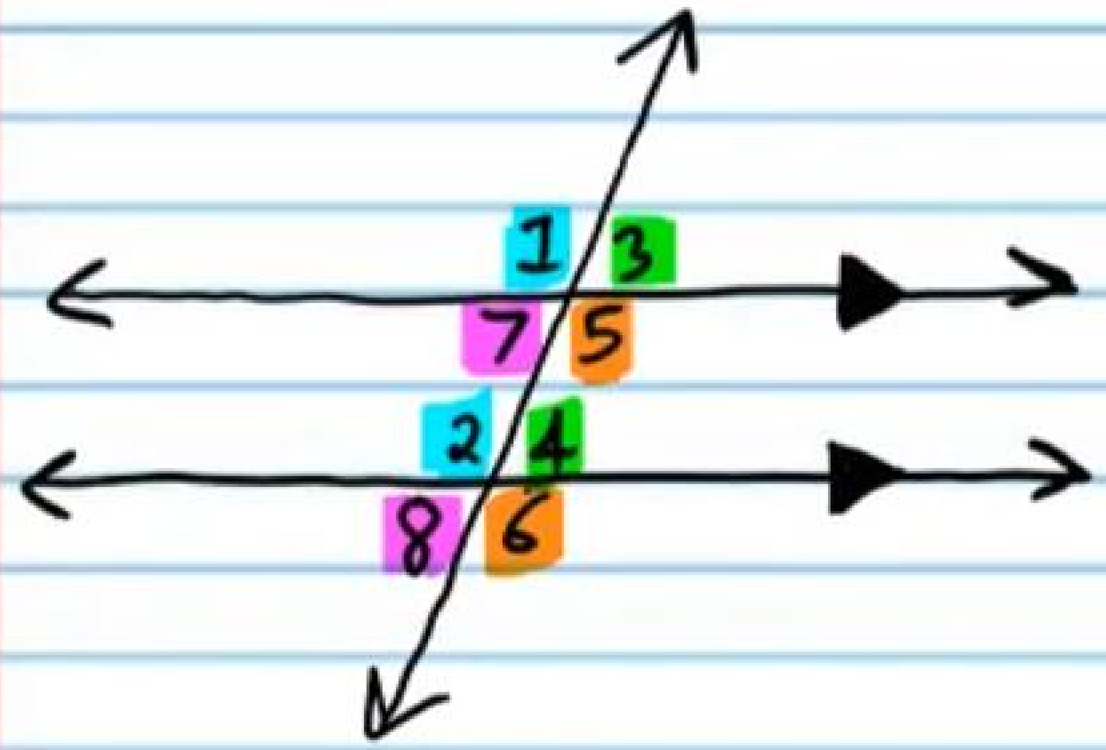
$\boxed{\text{Vert. } \angle\text{'s}}$

⑫


$\boxed{\text{Alt. Ext. } \angle\text{'s}}$

Transversals Through Parallel Lines

Corresponding Angles Postulate: If parallel lines are cut by a transversal, then corresponding angles formed are congruent.



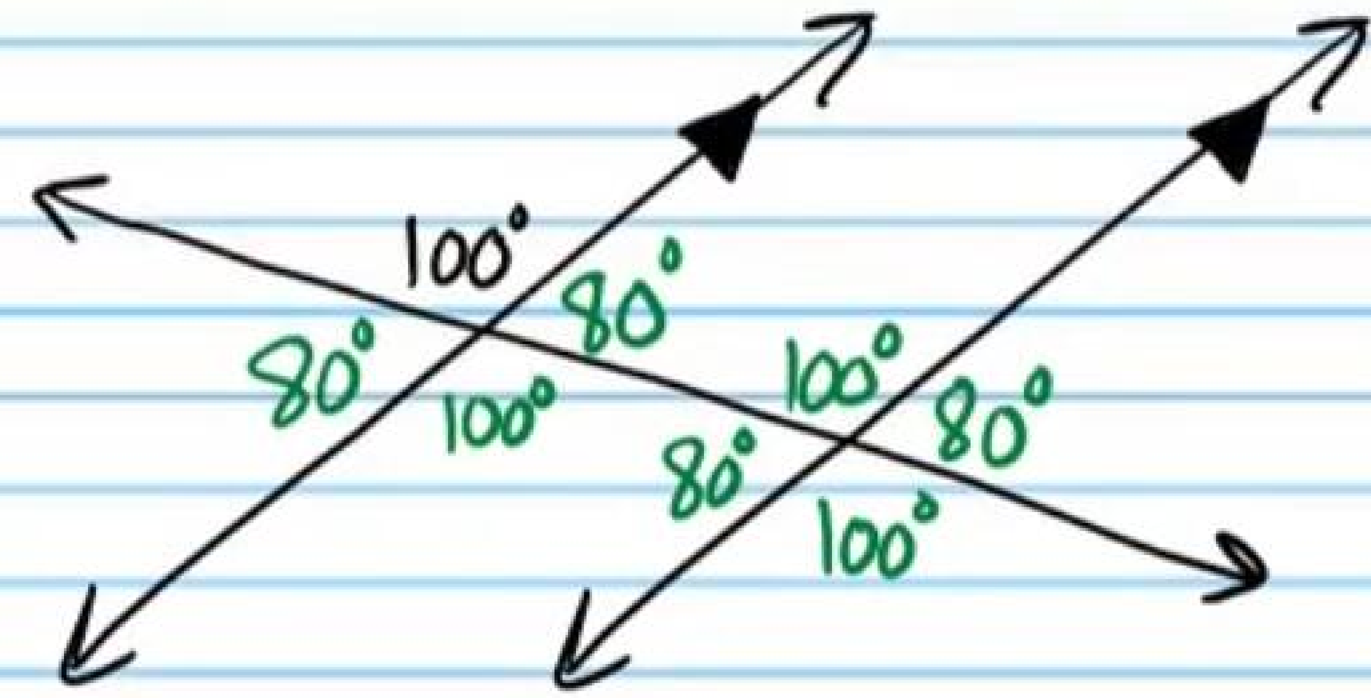
$\angle 1 \cong \angle 2$

$\angle 3 \cong \angle 4$

$\angle 5 \cong \angle 6$

$\angle 7 \cong \angle 8$

Find all angles formed by the transversals below.

⑬


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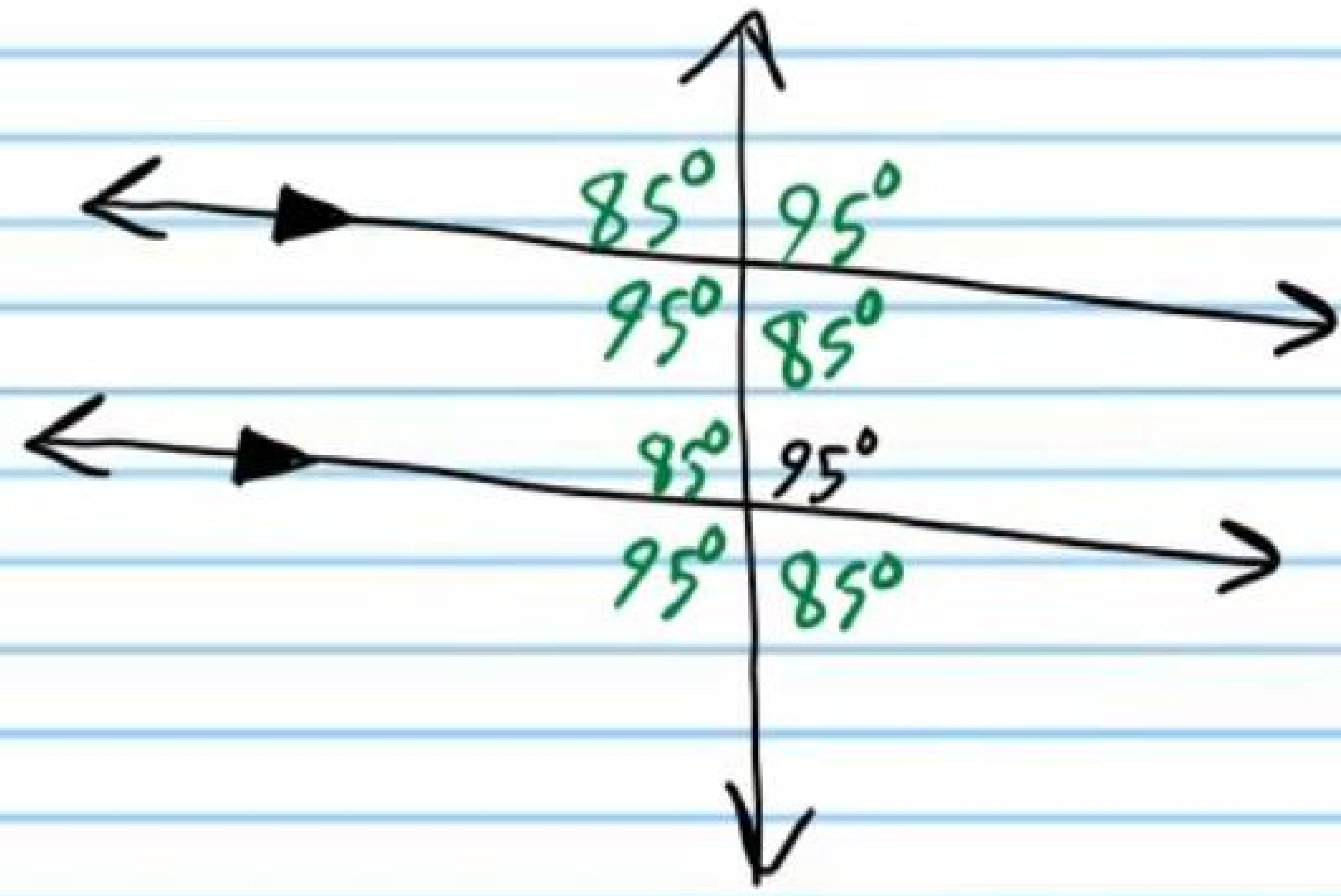
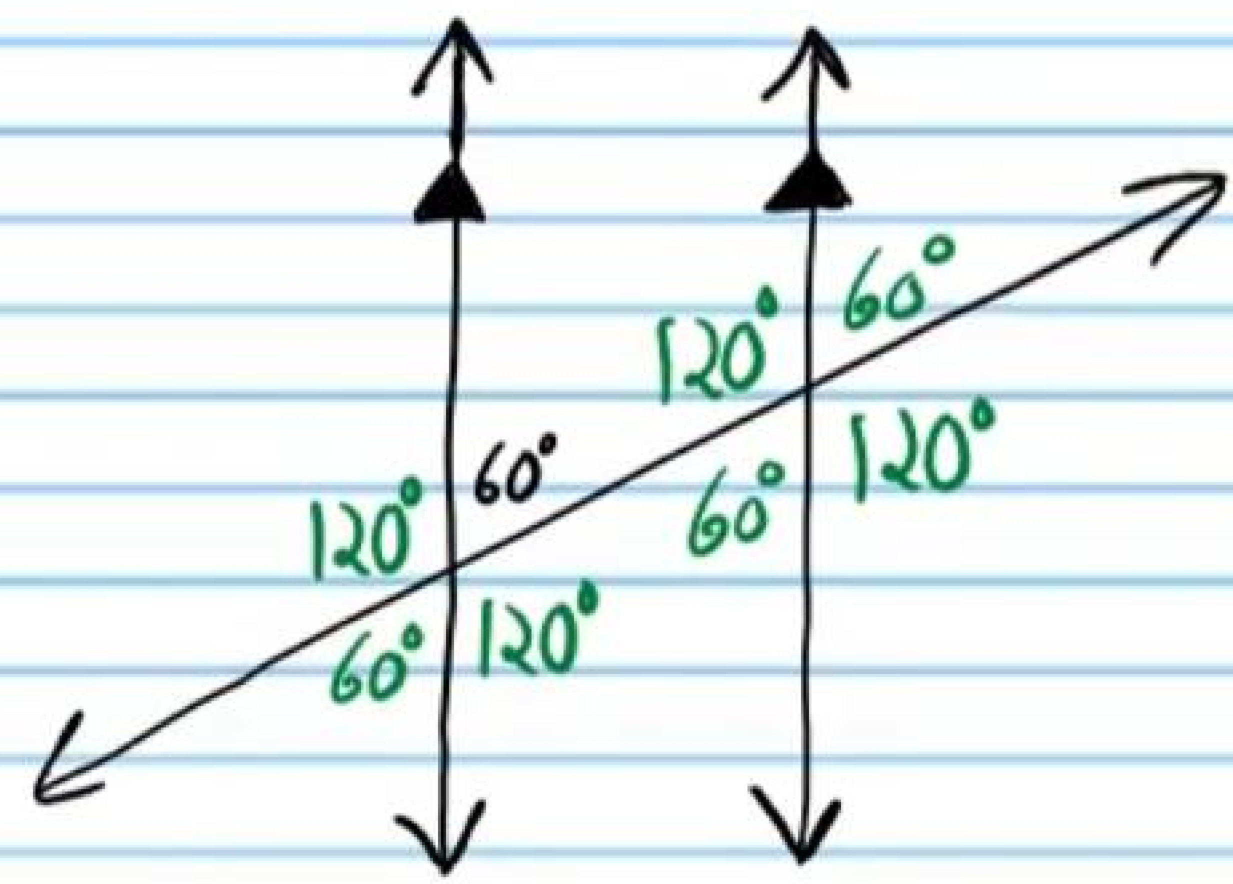
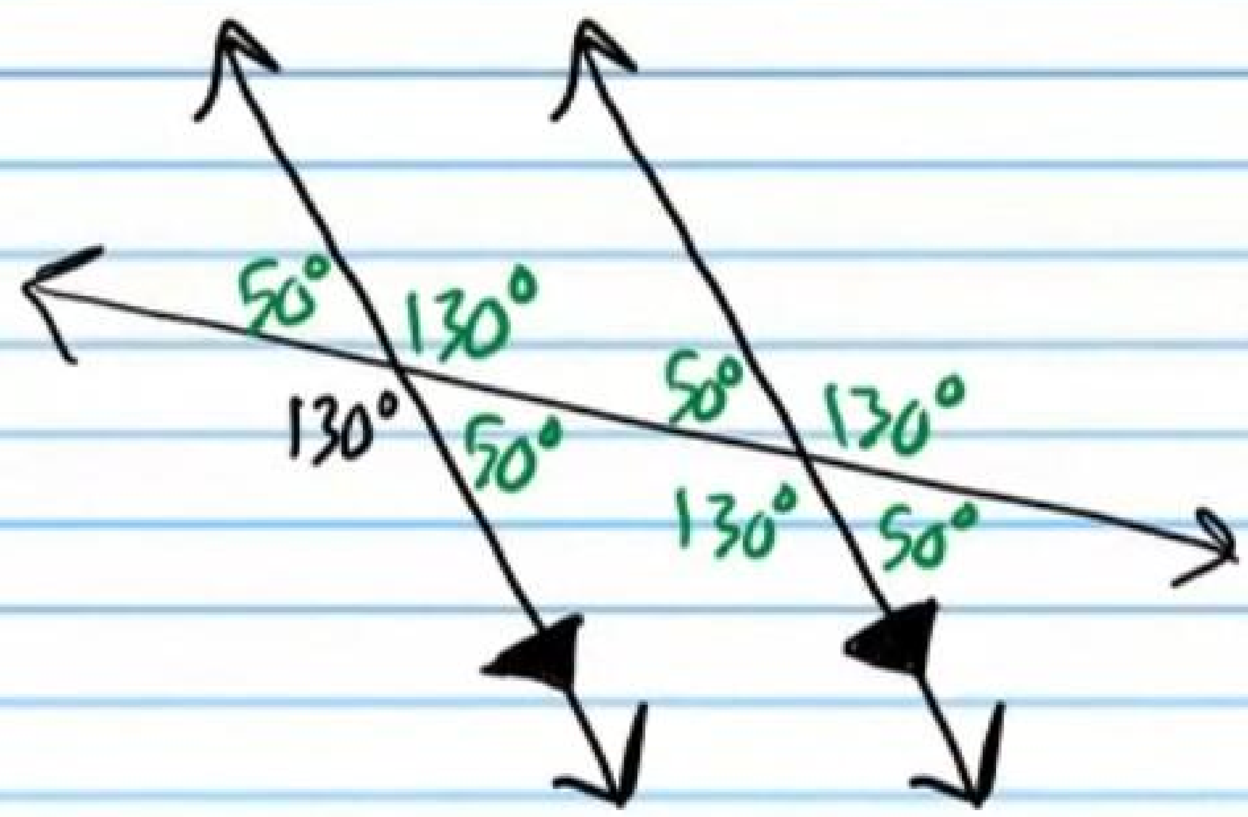
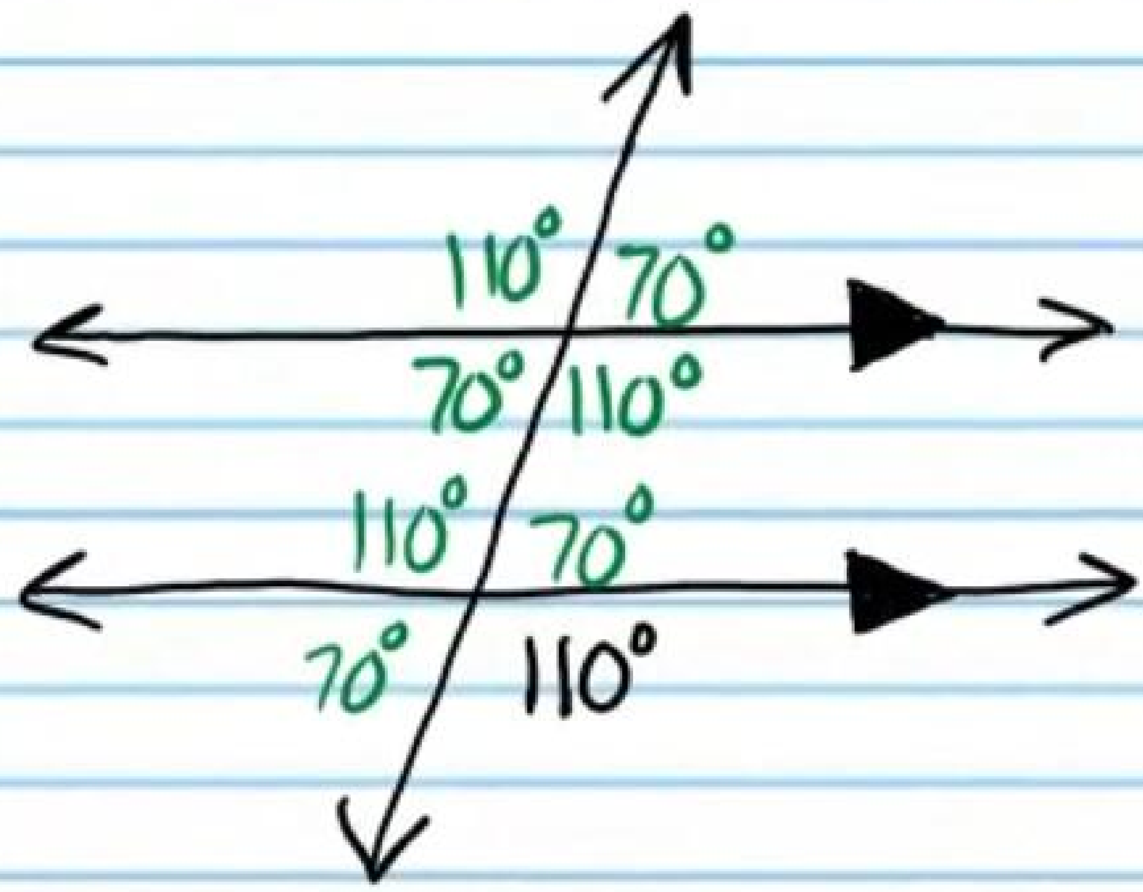


⑮


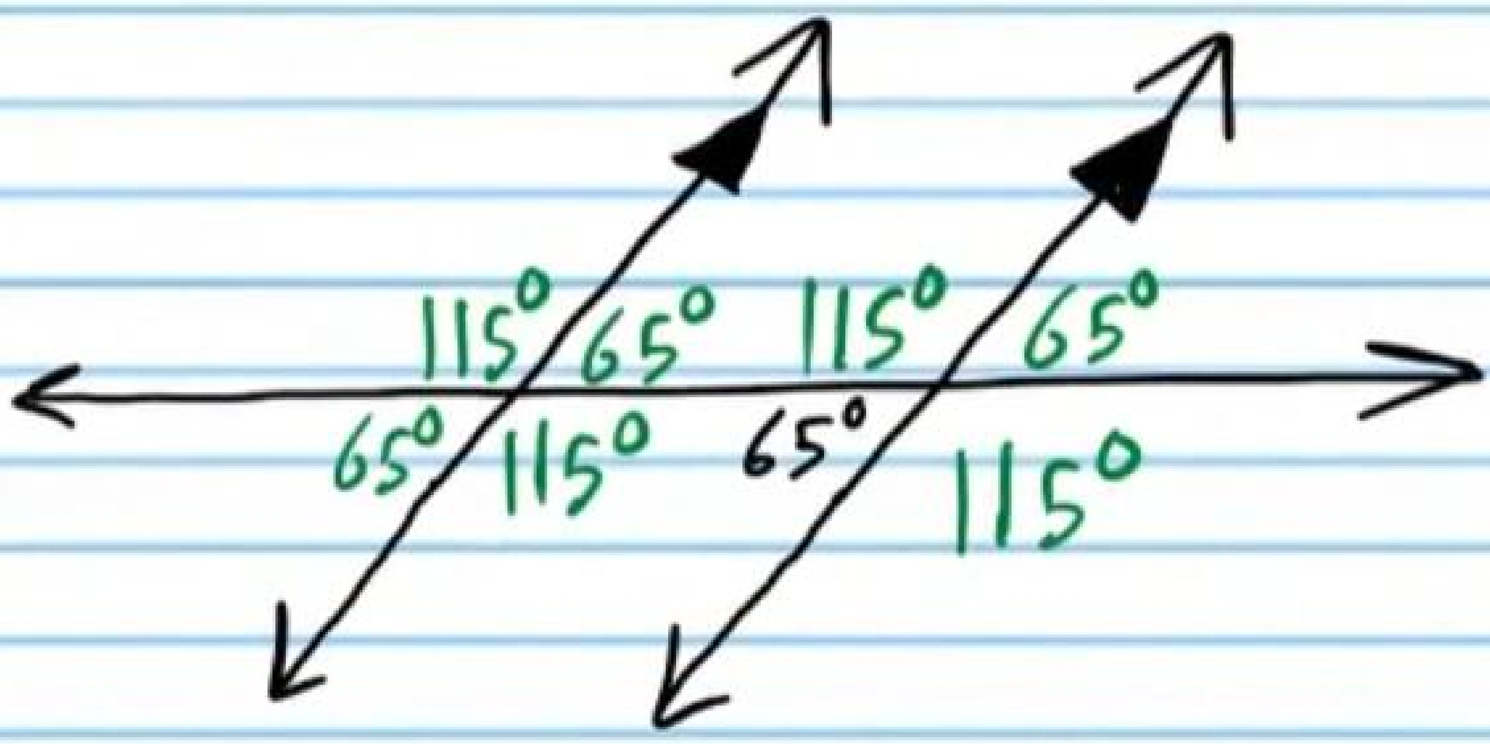
(16)



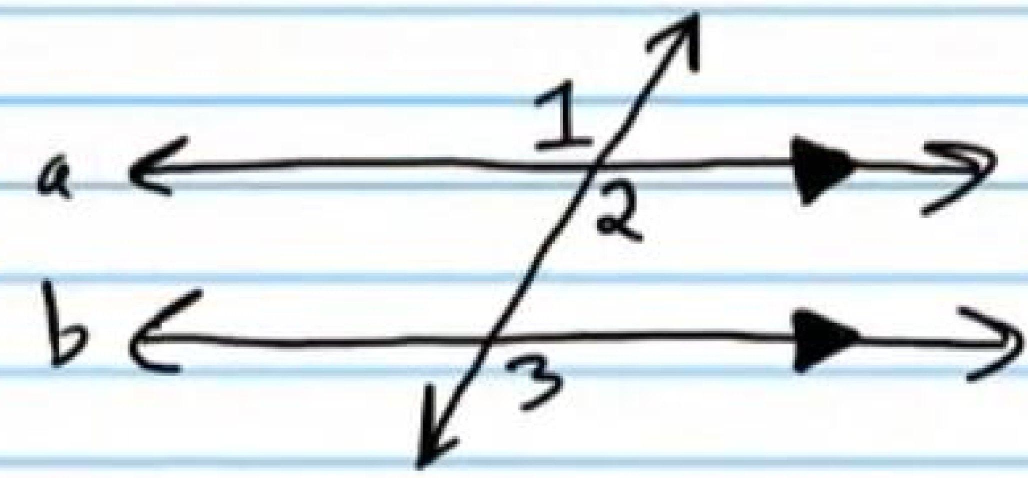
(17)



(18)

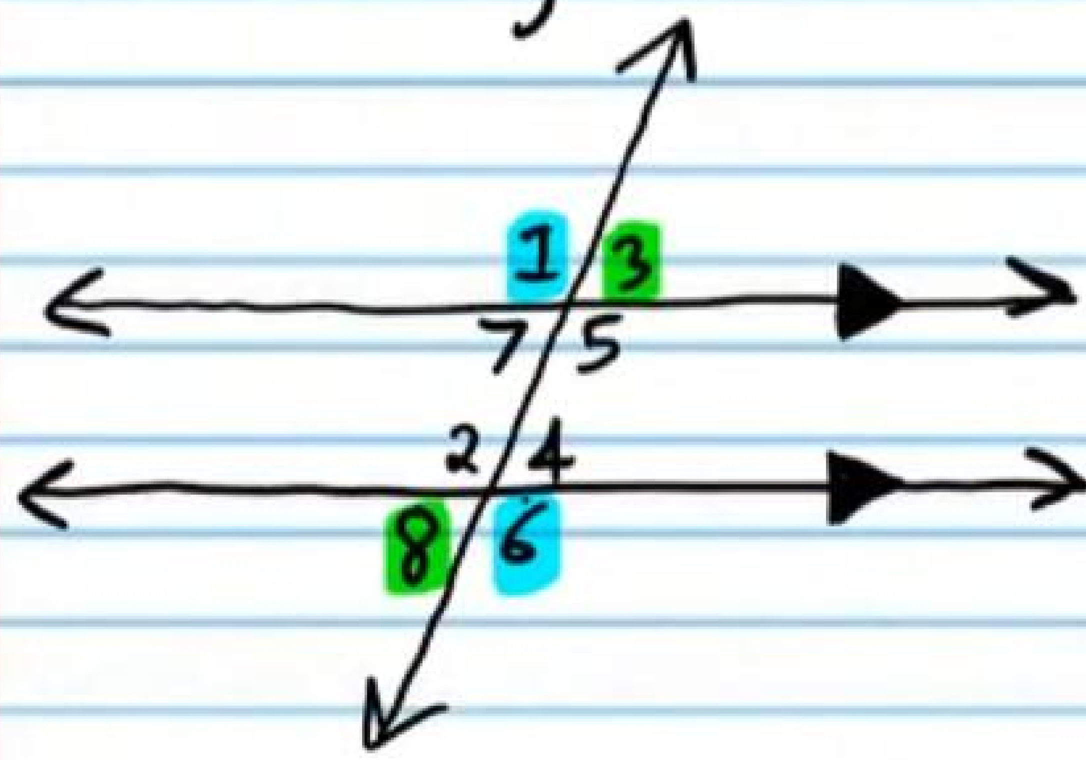


(19) Use the Corresponding Angles Postulate in a two-column proof to show that if two lines cut by a transversal are parallel, then alternate exterior angles are congruent.



Statement	Reason
I) $a \parallel b$	Given.
II) $m\angle 3 = m\angle 2$	Corr. ∠'s Post.
III) $m\angle 1 = m\angle 2$	Vert. ∠'s Thm.
IV) $m\angle 1 = m\angle 3$	Trans. Prop of =.

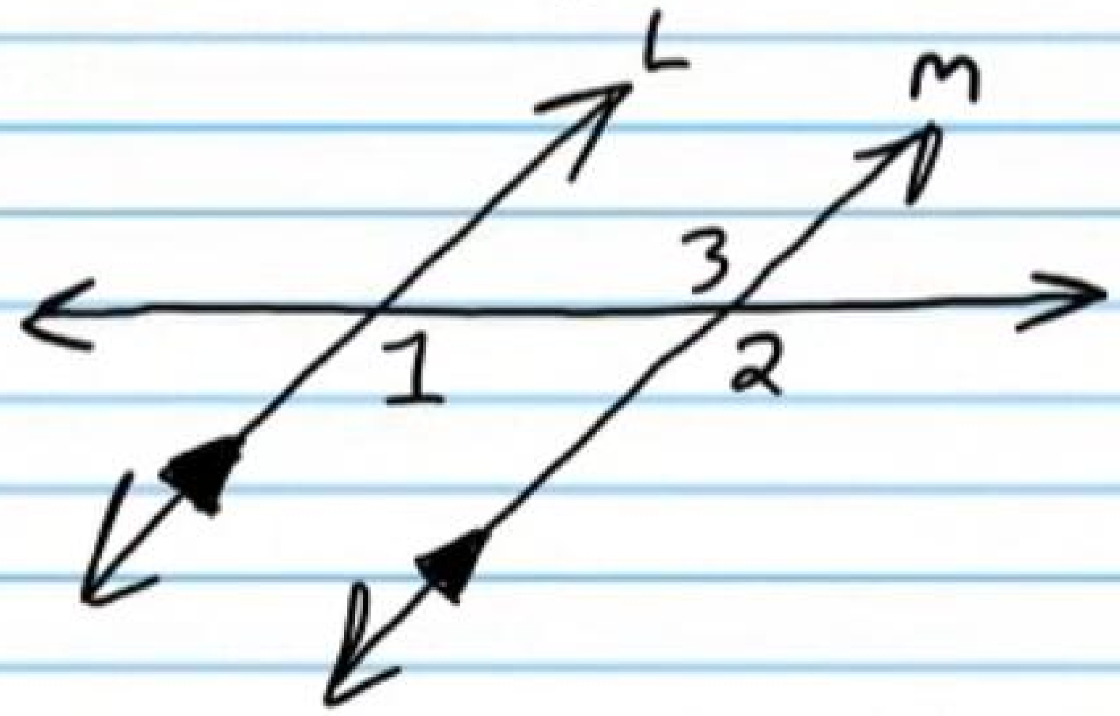
Alternate Exterior Angles Theorem: If parallel lines are cut by a transversal, then alternate exterior angles formed are congruent.



$\angle 1 \cong \angle 6$

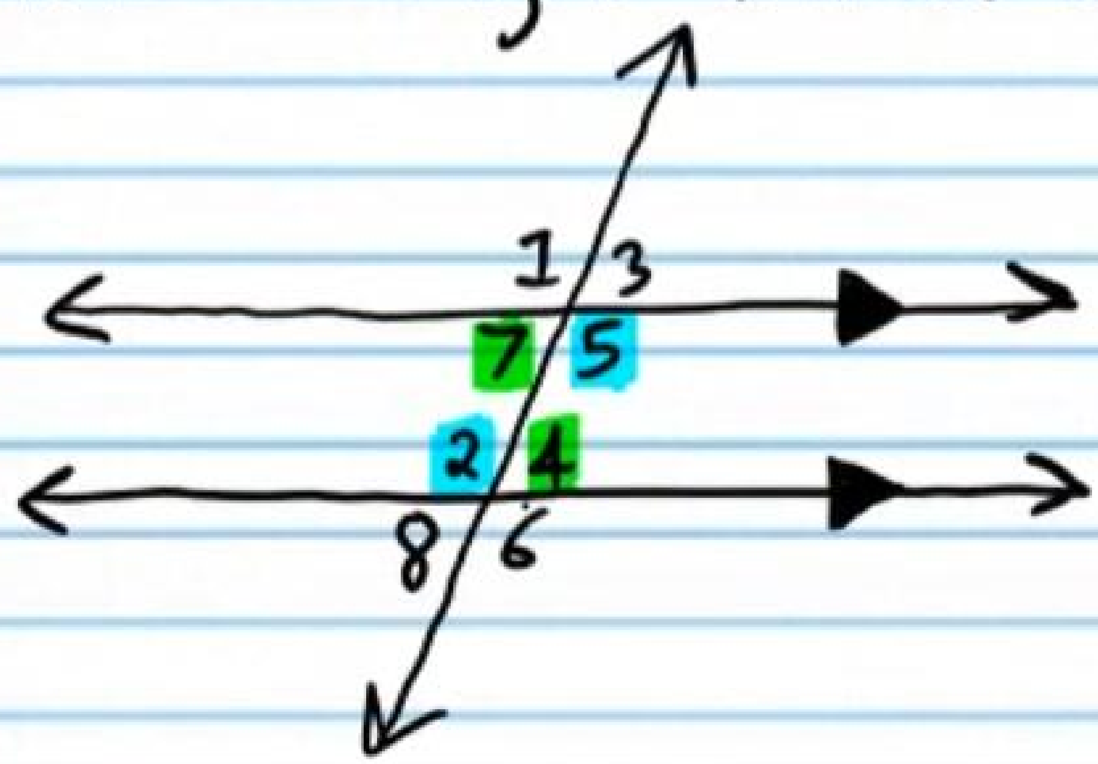
$\angle 3 \cong \angle 8$

(20) Use the Corresponding Angles Postulate in a two-column proof to show that if two lines cut by a transversal are parallel, then alternate interior angles are congruent.



Statement	Reason
I) $L \parallel m$	Given.
II) $m\angle 1 = m\angle 2$	Corr. ∠'s Post.
III) $m\angle 3 = m\angle 2$	Vert. ∠'s Thm.
IV) $m\angle 1 = m\angle 3$	Trans. Prop of =.

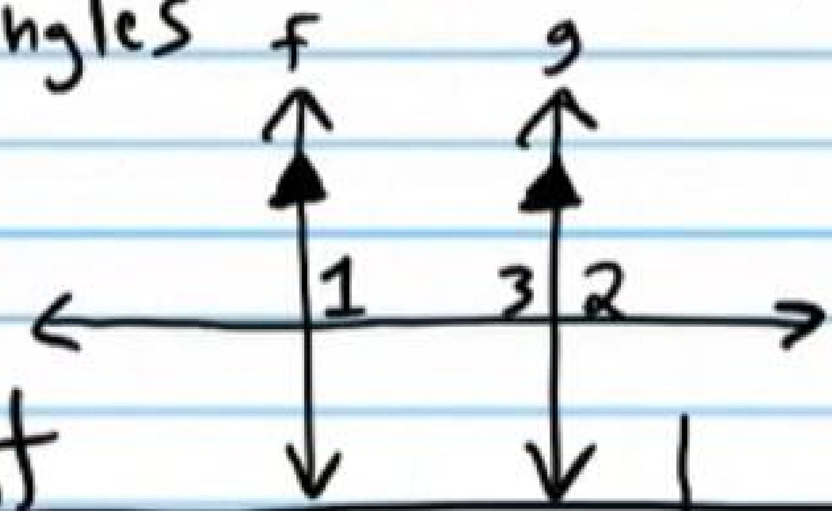
Alternate Interior Angles Theorem: If parallel lines are cut by a transversal, then alternate interior angles formed are congruent.



$\angle 7 \cong \angle 4$

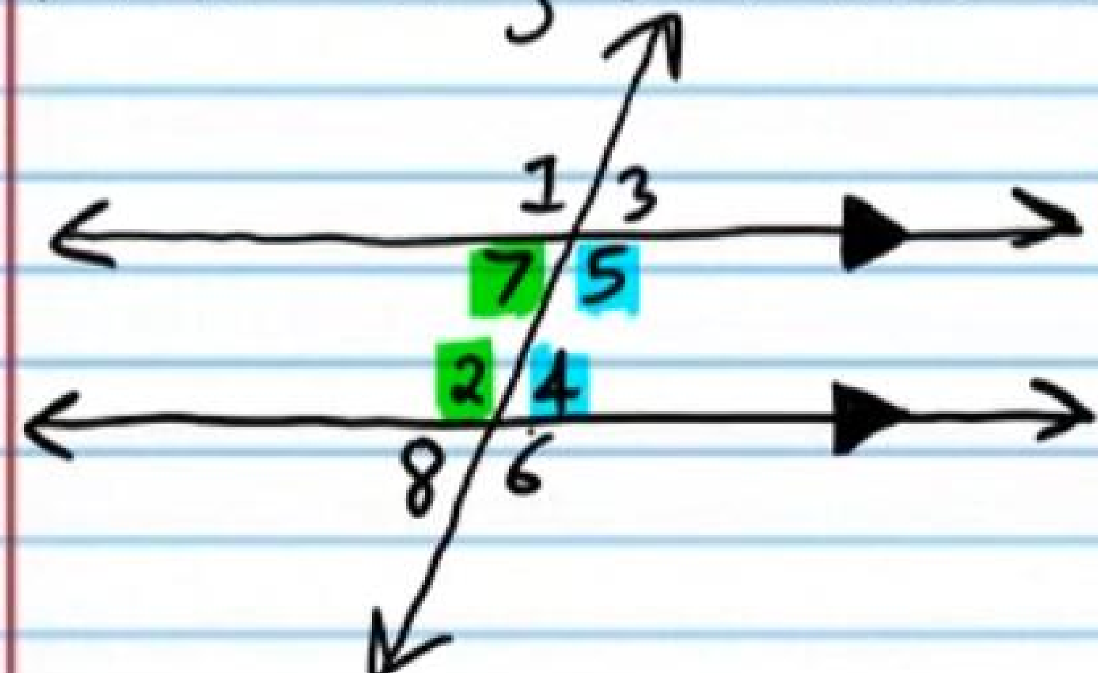
$\angle 2 \cong \angle 5$

(21) Use the Corresponding Angles Postulate and the Linear Pair Postulate in a two-column proof to show that if two lines cut by a transversal are parallel, then consecutive interior angles are supplementary.



Statement	Reason
I) $f \parallel g$	Given.
II) $m\angle 1 = m\angle 2$	Corr. ∠'s Post.
III) ∠3 & ∠2 form a linear pair.	Given (diagram)
IV) ∠3 & ∠2 are supp. ∠'s.	Linear Pair Post.
V) $m\angle 3 + m\angle 2 = 180$	Def of supp ∠'s.
VI) $m\angle 3 + m\angle 1 = 180$	Subs. Prop of =.
VII) ∠3 & ∠1 are supp ∠'s	Def. of supp ∠'s.

Consecutive Interior Angles Theorem: If parallel lines are cut by a transversal, then consecutive interior angles formed are supplementary.

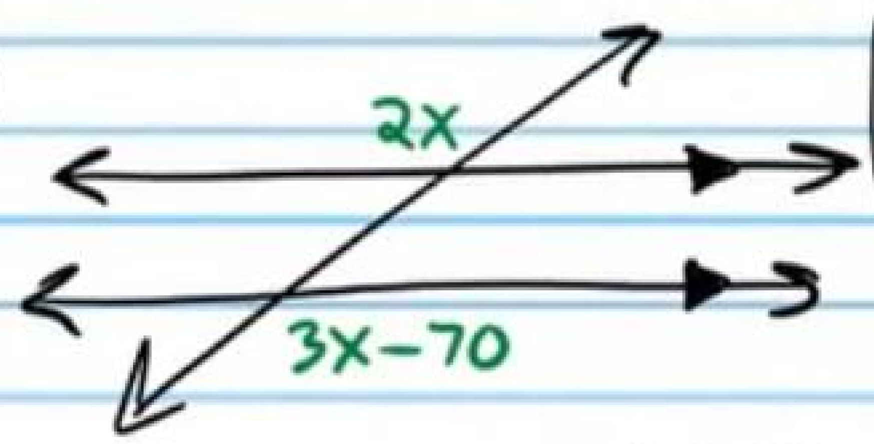


∠7 & ∠2 are supplementary.

∠5 & ∠4 are supplementary.

Find x and state the theorem(s) and/or postulate(s) used. You don't have to do a complete proof.

(22)

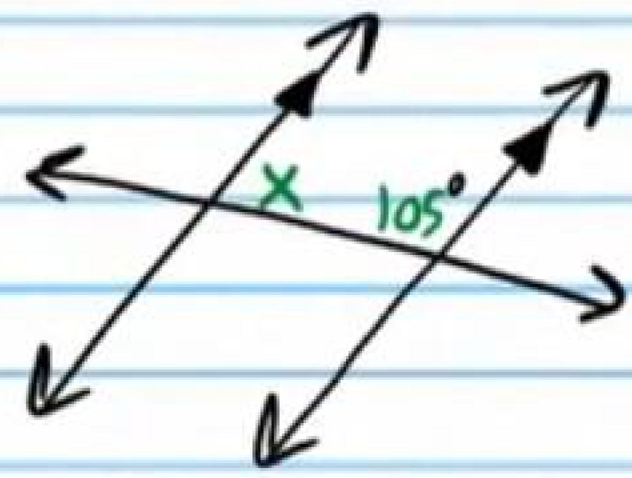


$2x = 3x - 70$ $\boxed{\text{AEAT}}$

$0 = x - 70$

$\boxed{70 = x}$

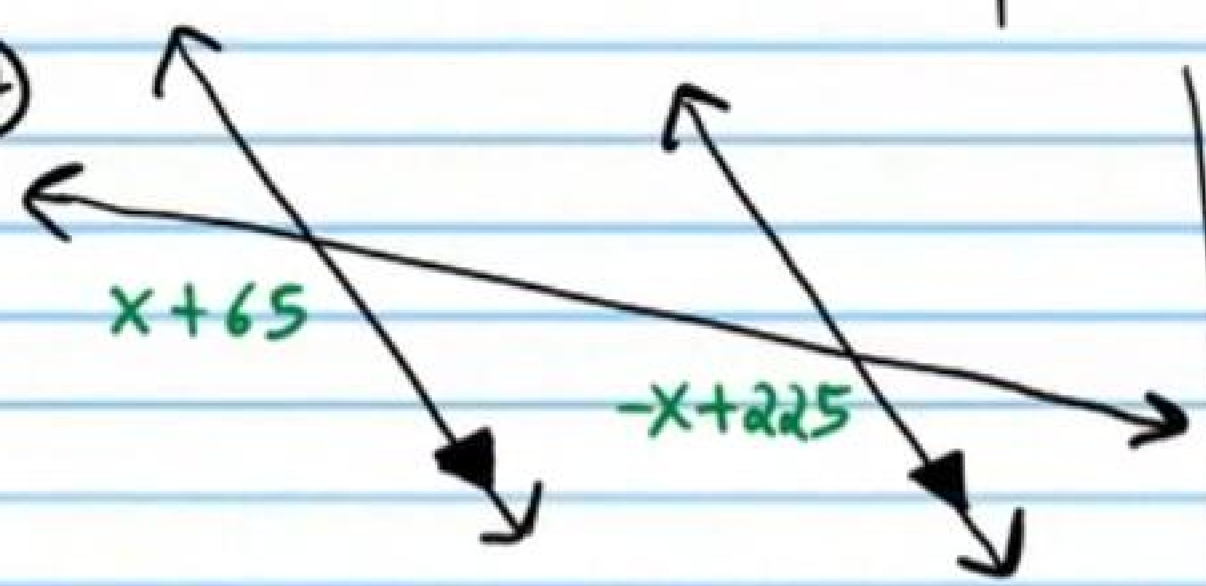
(23)



$x + 105 = 180$ $\boxed{\text{CIAT}}$

$\boxed{x = 75^\circ}$

(24)



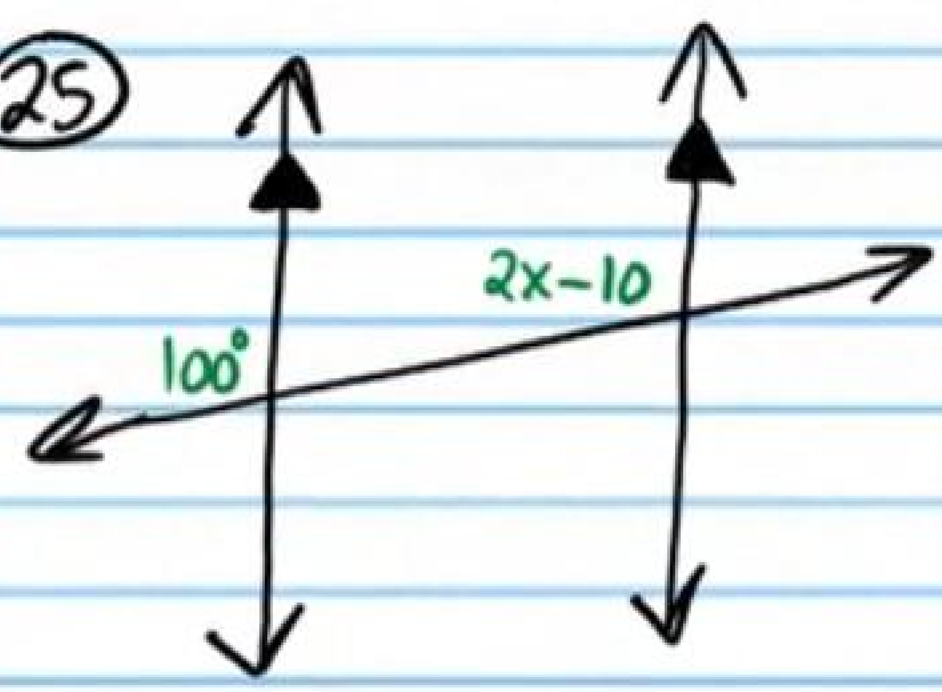
$x + 65 = -x + 225$ $\boxed{\text{CAP}}$

$2x + 65 = 225$

$2x = 160$

$\boxed{x = 80}$

(25)

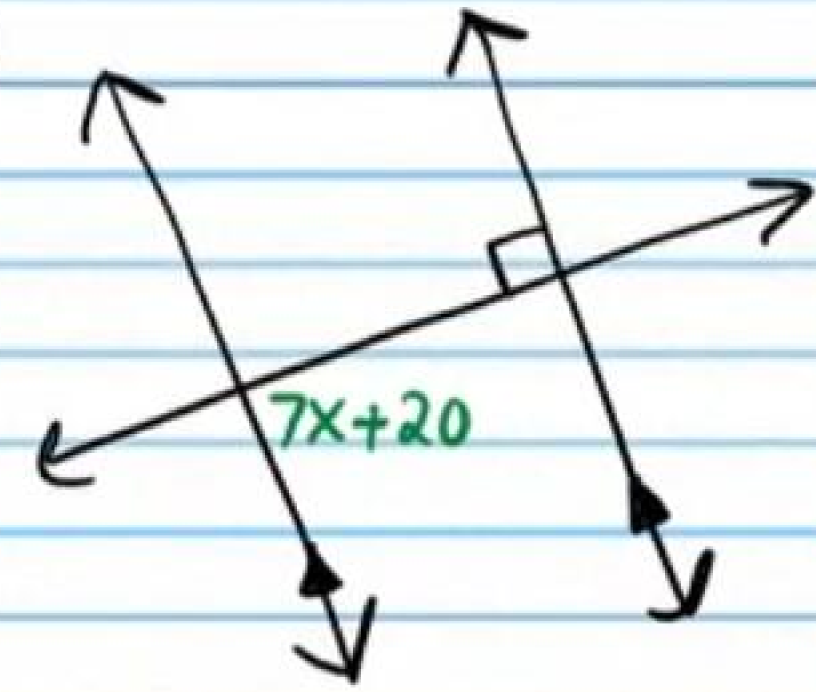


$100 = 2x - 10$ $\boxed{\text{CAP}}$

$110 = 2x$

$\boxed{55 = x}$

(26)

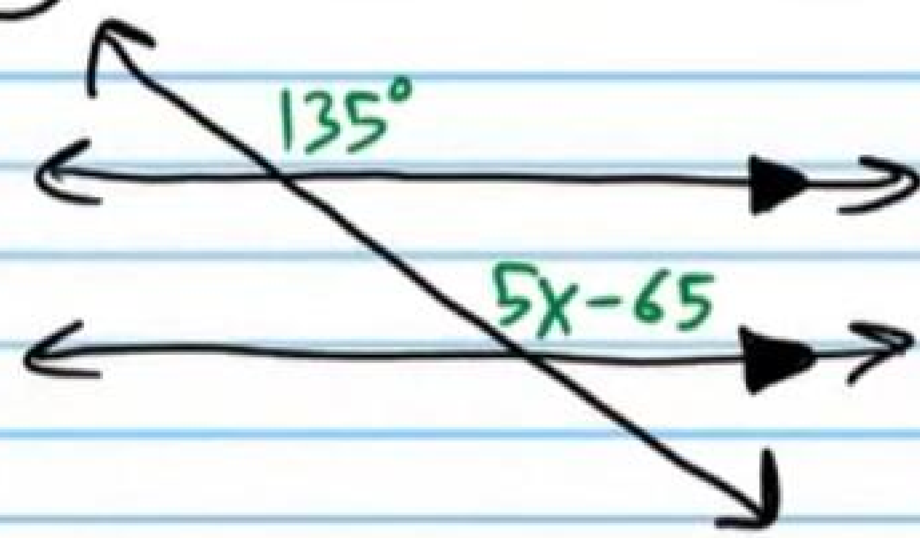


$7x + 20 = 90^\circ$ $\boxed{\text{AIAT}}$

$7x = 70$

$\boxed{x = 10}$

(27)

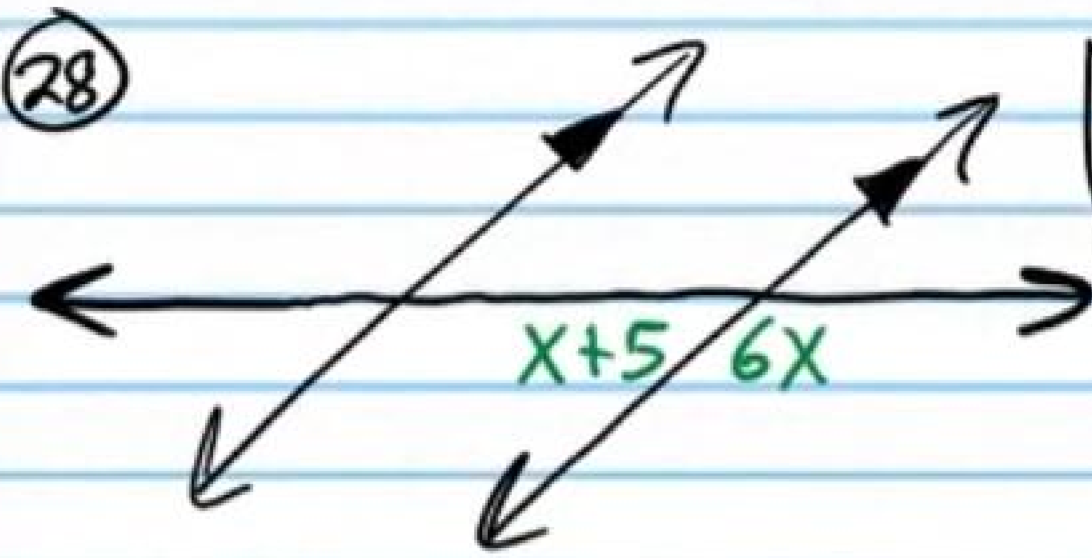


$135 = 5x - 65$ $\boxed{\text{CAP}}$

$200 = 5x$

$\boxed{40 = x}$

(28)



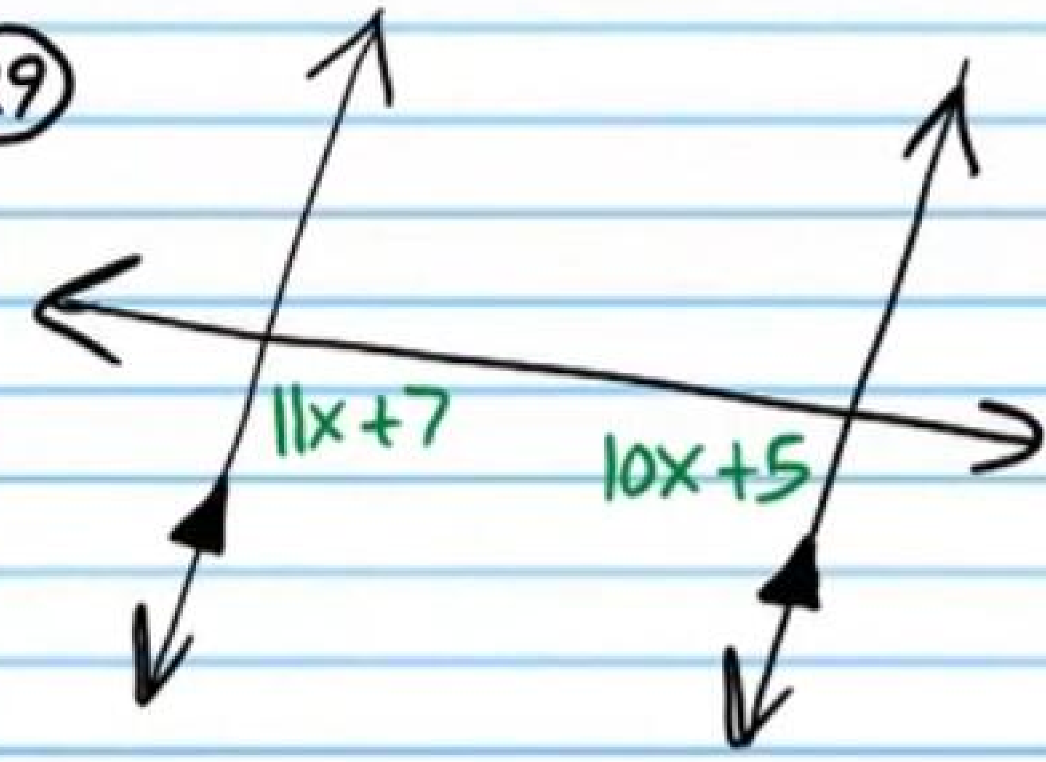
$x + 5 + 6x = 180$ $\boxed{\text{LPP}}$

$7x + 5 = 180$

$7x = 175$

$\boxed{x = 25}$

(29)



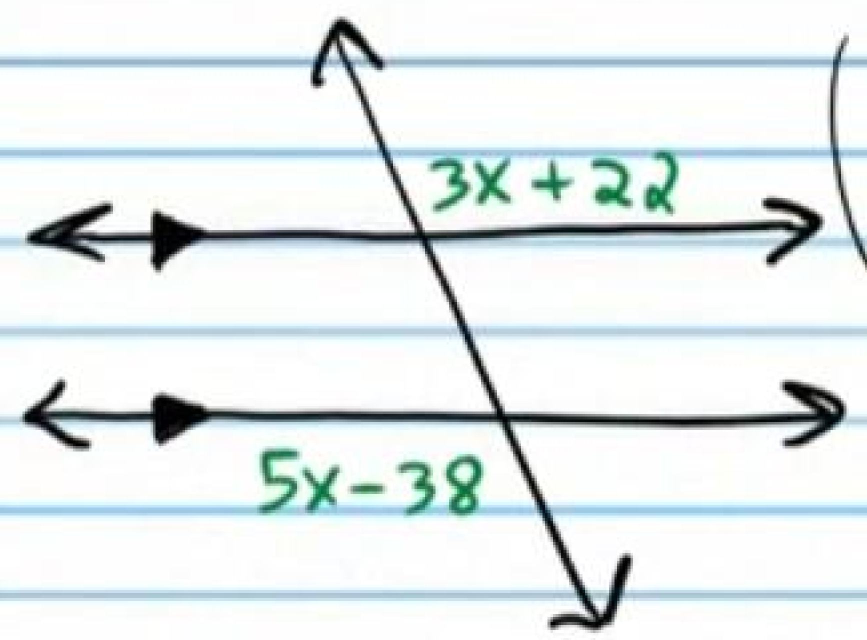
$11x + 7 + 10x + 5 = 180$ $\boxed{\text{CIAT}}$

$21x + 12 = 180$

$21x = 168$

$\boxed{x = 8}$

(30)



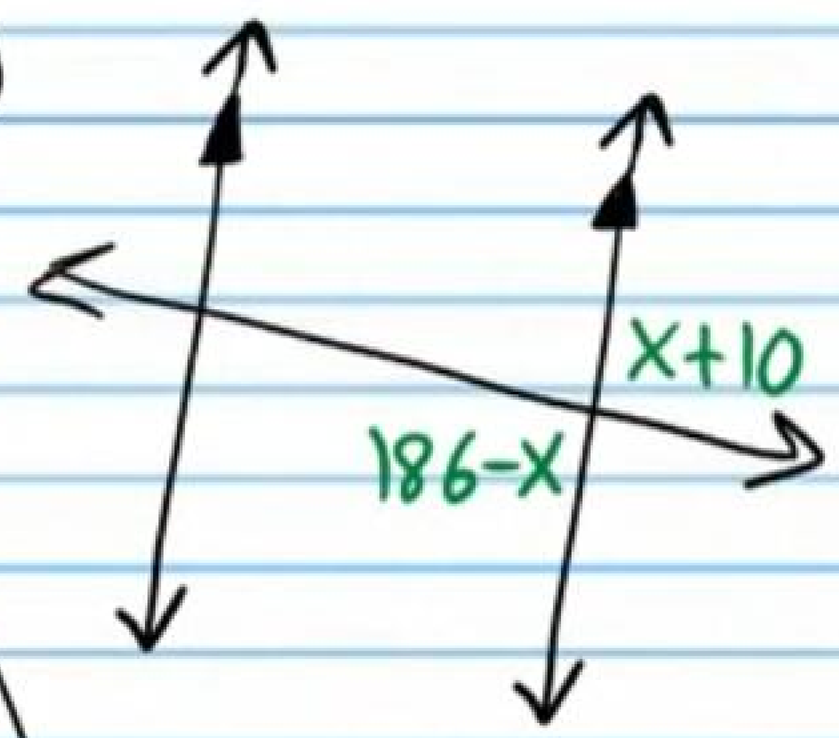
$3x+22=5x-38$ $\boxed{\text{AEAT}}$

$22=2x-38$

$60=2x$

$\boxed{30=x}$

* (31)



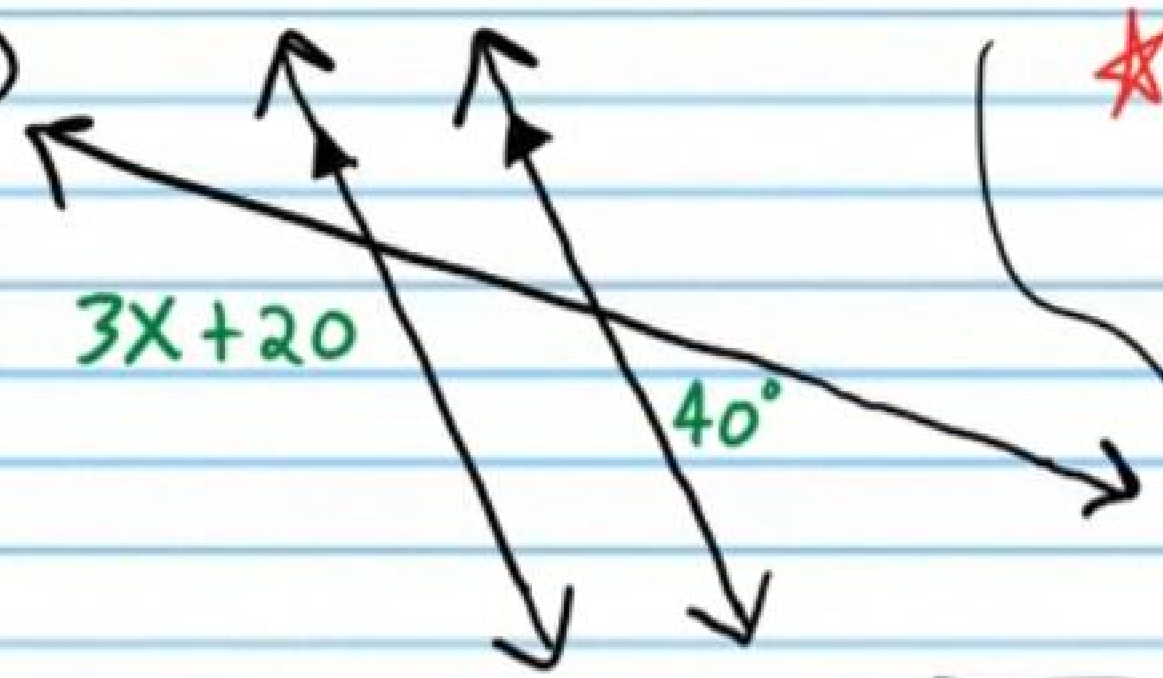
$186-x=x+10$ $\boxed{\text{VAT}}$

$186=2x+10$

$176=2x$

$\boxed{88=x}$

* (32)



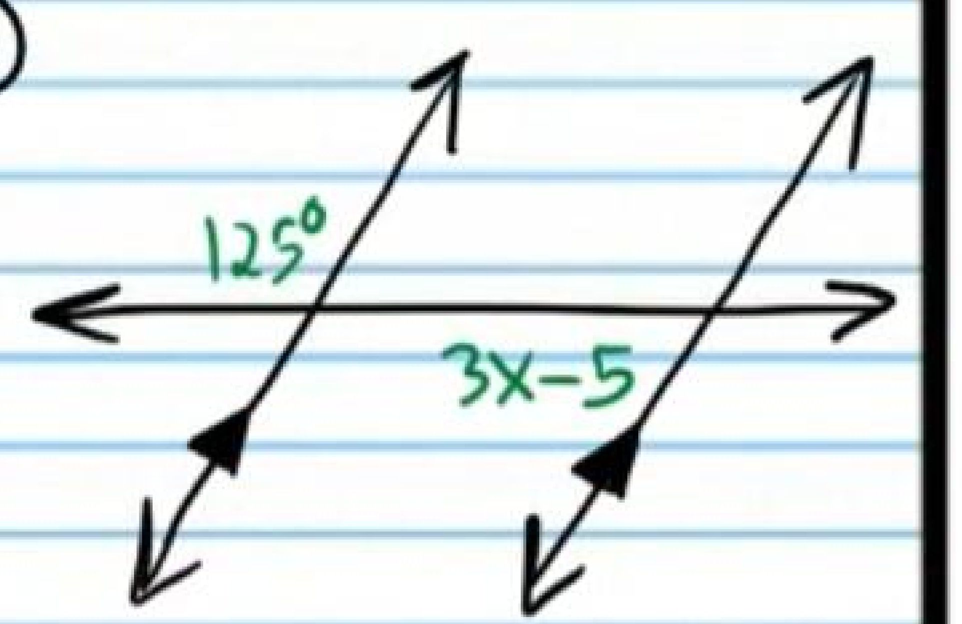
$3x+20+40=180$ $\boxed{\text{N/A}}$

$3x+60=180$

$3x=120$

$\boxed{x=40}$

* (33)



$125+3x-5=180$ $\boxed{\text{N/A}}$

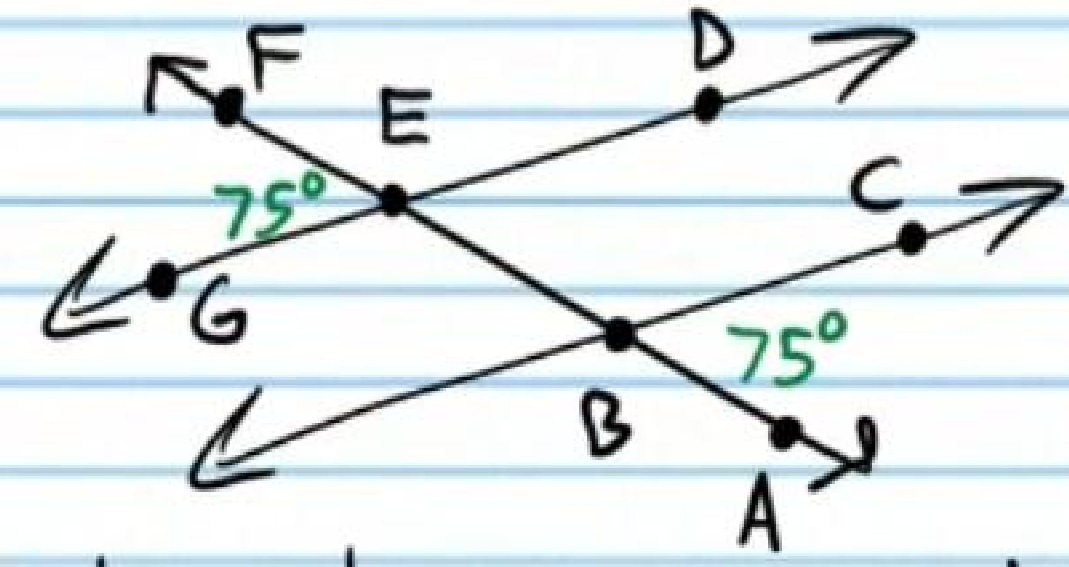
$120+3x=180$

$3x=60$

$\boxed{x=20}$

Corresponding Angles Converse Postulate: If corresponding angles formed by a transversal are congruent, then the lines cut by the transversal are parallel.

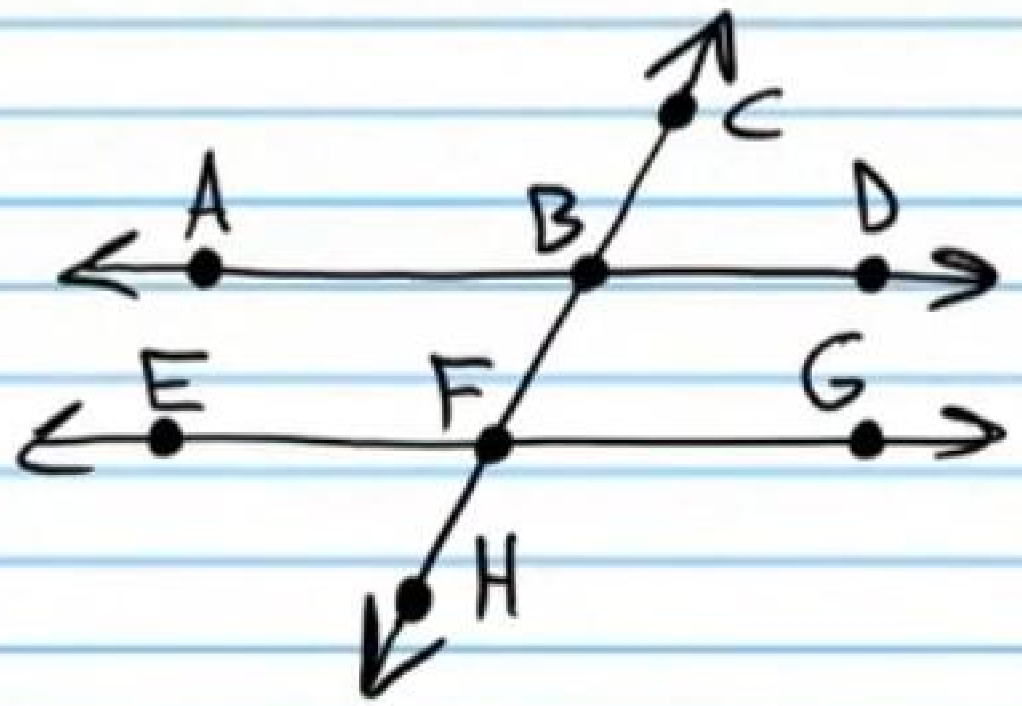
(34) Use a two-column proof and the Corresponding Angles Converse Postulate to show that the lines below are parallel.



Statement	Reason
I) $m\angle FEG = 75°$ & $m\angle ABC = 75°$	Given (diagram).
II) $m\angle FEG = m\angle DEB$	Vert. ∠'s Thm.
III) $75° = m\angle DEB$	Trans. Prop of =.
IV) $m\angle ABC = m\angle DEB$	Trans. Prop of =
V) $\overleftrightarrow{GD} \parallel \overleftrightarrow{BC}$	Corr ∠'s Conv. Post.

Alternate Exterior Angles Converse Theorem: If alternate exterior angles formed by a transversal are congruent, then the lines cut by the transversal are parallel.

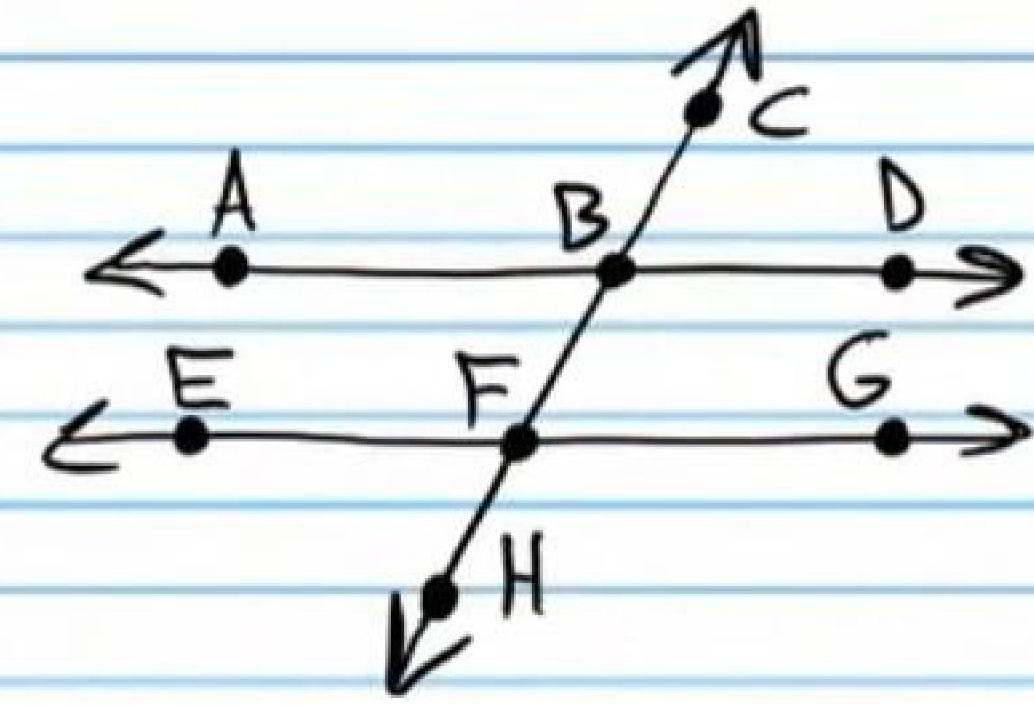
(35) Use a two-column proof and the Corresponding Angles Converse Postulate to prove the Alternate Exterior Angles Converse Theorem.



Statement	Reason
I) $m\angle ABC = m\angle HFG$	Given.
II) $m\angle ABC = m\angle FBD$	Vert. ∠'s Thm.
III) $m\angle HFG = m\angle FBD$	Trans. Prop of =.
IV) $\overleftrightarrow{AD} \parallel \overleftrightarrow{EG}$	Corr. ∠'s Conv Post.

Alternate Interior Angles Converse Theorem: If alternate interior angles formed by a transversal are congruent, then the lines cut by the transversal are parallel.

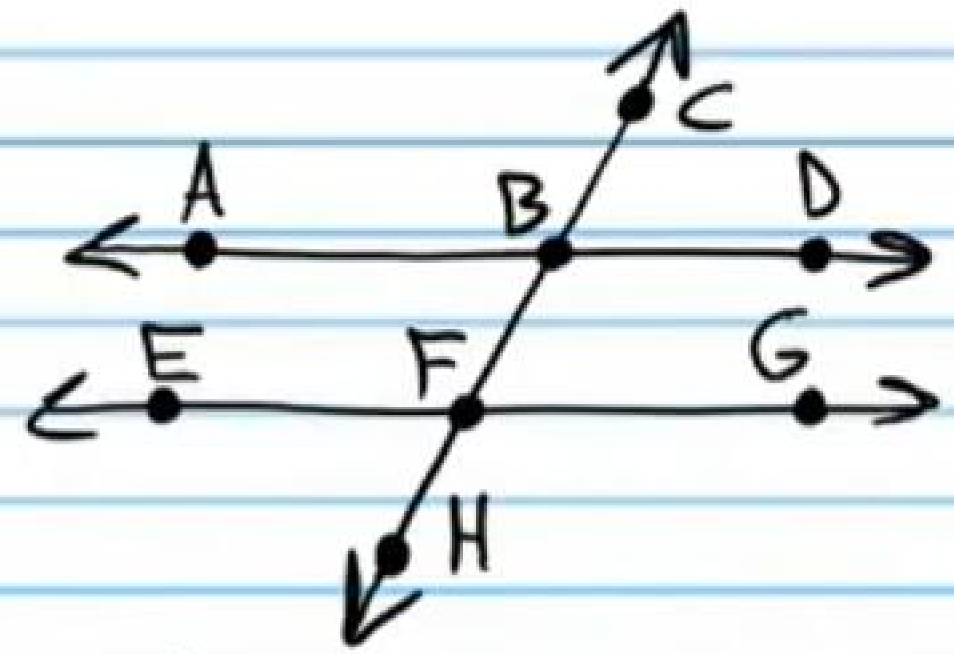
(36) Use a two-column proof and the Corresponding Angles Converse Postulate to prove the Alternate Interior Angles Converse Theorem.



Statement	Reason
I) $m\angle ABF = m\angle BFG$	Given.
II) $m\angle ABF = m\angle CBD$	Vert. ∠'s Thm.
III) $m\angle CBD = m\angle BFG$	Trans. Prop of =.
IV) $\overleftrightarrow{AB} \parallel \overleftrightarrow{EG}$	Corr. ∠'s Conv Post.

Consecutive Interior Angles Converse Theorem: If consecutive interior angles formed by a transversal are supplementary, then the lines cut by the transversal are parallel.

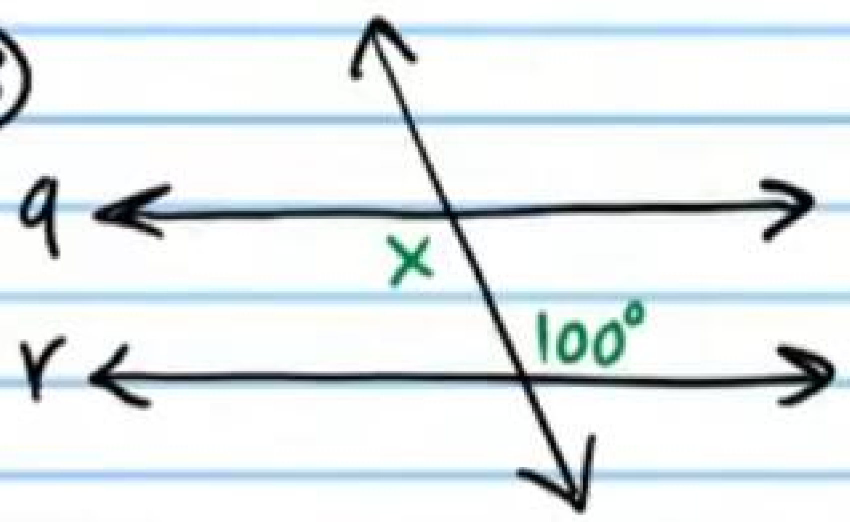
(37) Use a two-column proof and the Corresponding Angles Converse Postulate to prove the Consecutive Interior Angles Converse Theorem.



Statement	Reason
I) $\angle BFG$ & $\angle FBD$ are supp. $\angle$'s	Given.
II) $\angle CBD$ & $\angle FBD$ form a linear pair.	Diagram.
III) $\angle CBD$ & $\angle FBD$ are supp $\angle$'s	Linear Pair Post.
IV) $\angle BFG \cong \angle CBD$	$\cong$ Supp Thm.
V) $\overleftrightarrow{AD} \parallel \overleftrightarrow{EG}$	Corr $\angle$'s Conv. Post.

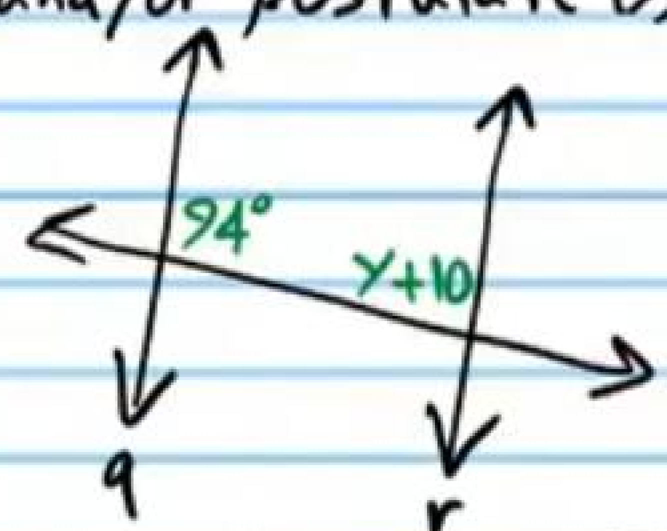
Find x and/or y such that lines q and r are parallel. You don't have to do a formal proof, but state the primary theorem(s) and/or postulate(s) used.

(38)



$\boxed{x = 100°}$ $\boxed{\text{AIACT}}$

(39)

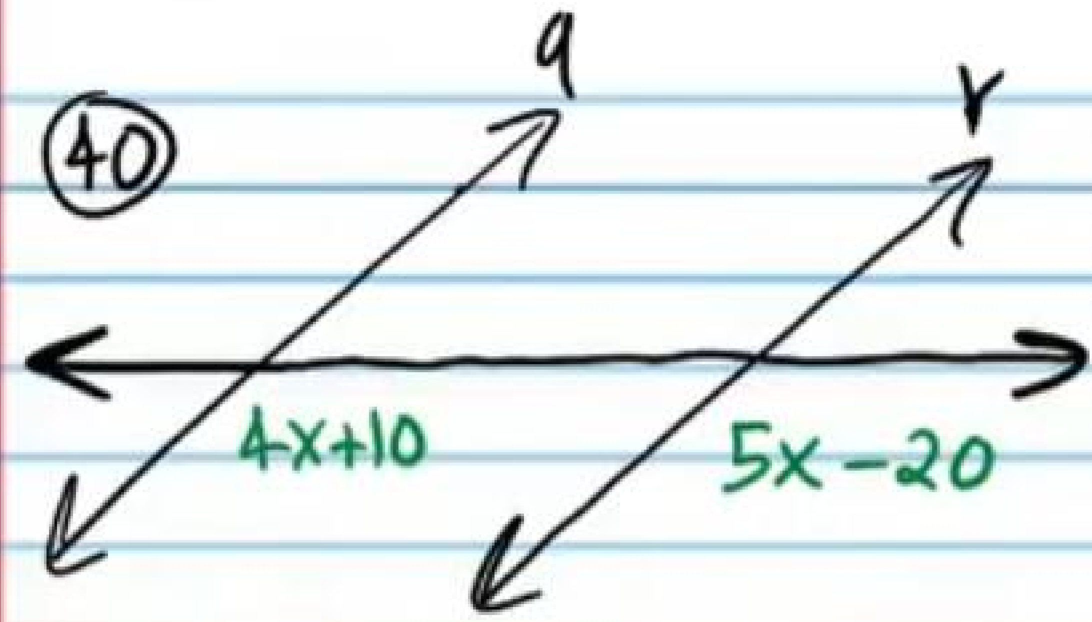


$(94) + (y+10) = 180$ $\boxed{\text{CIACT}}$

$y + 104 = 180$

$\boxed{y = 76}$

(40)

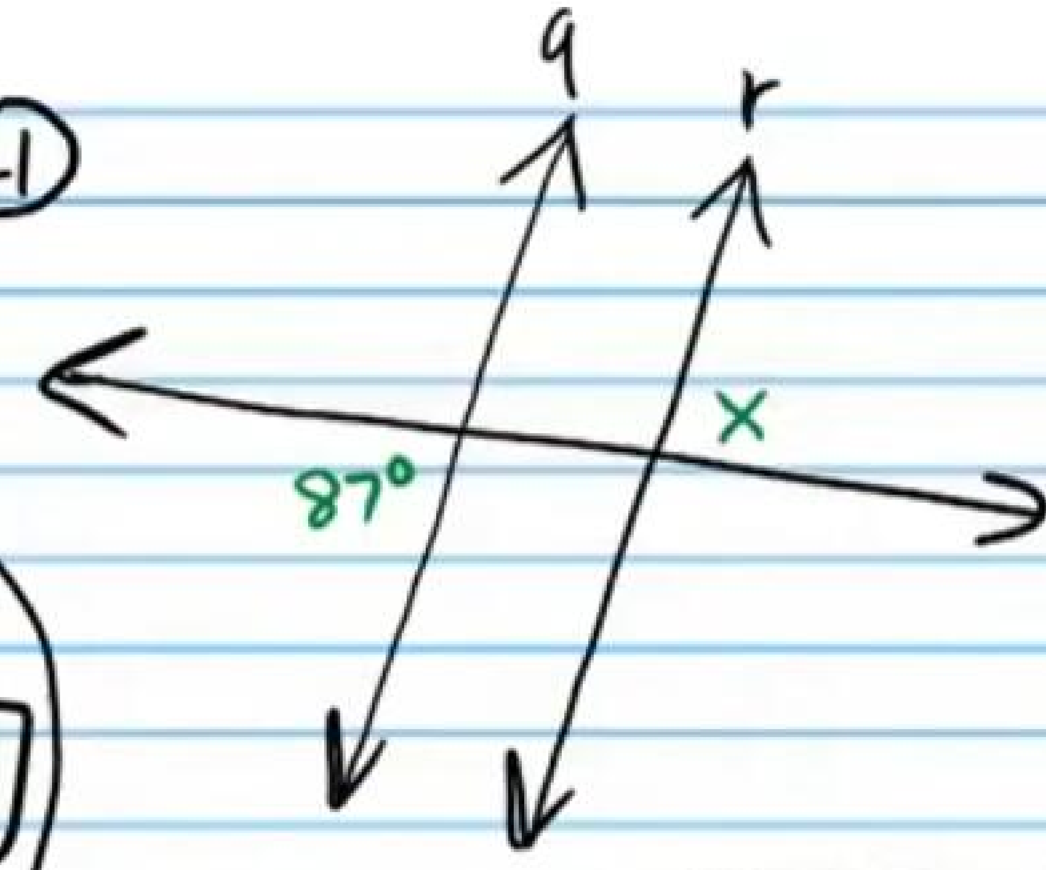


$4x + 10 = 5x - 20$ $\boxed{\text{CACP}}$

$10 = x - 20$

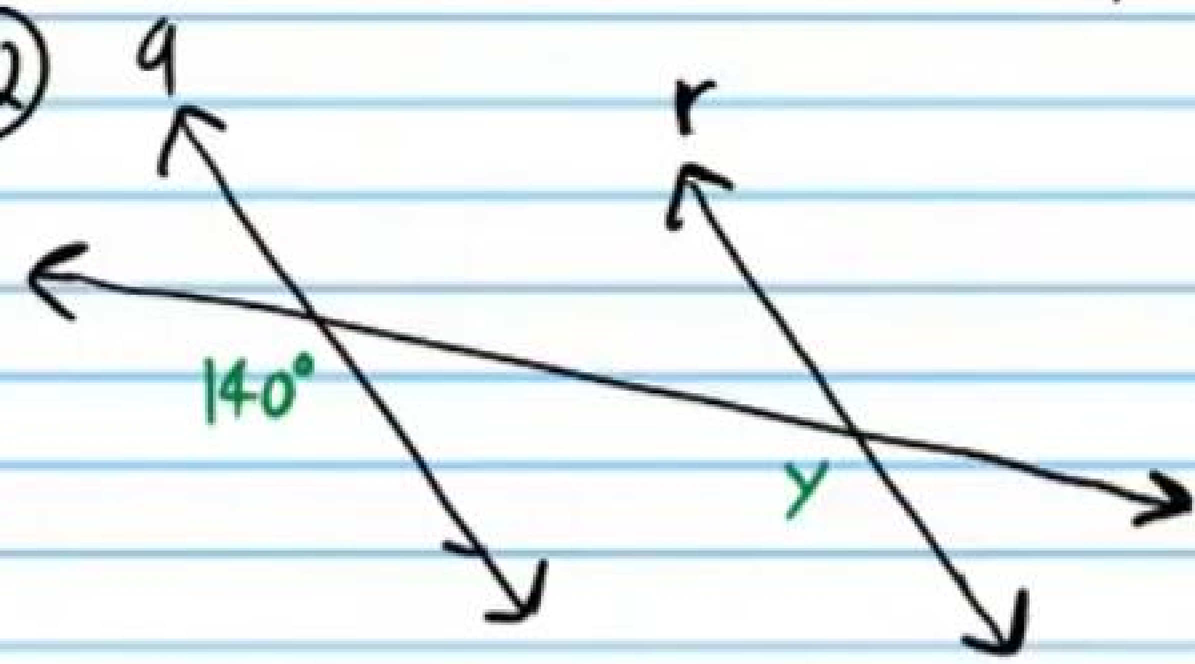
$\boxed{30 = x}$

(41)



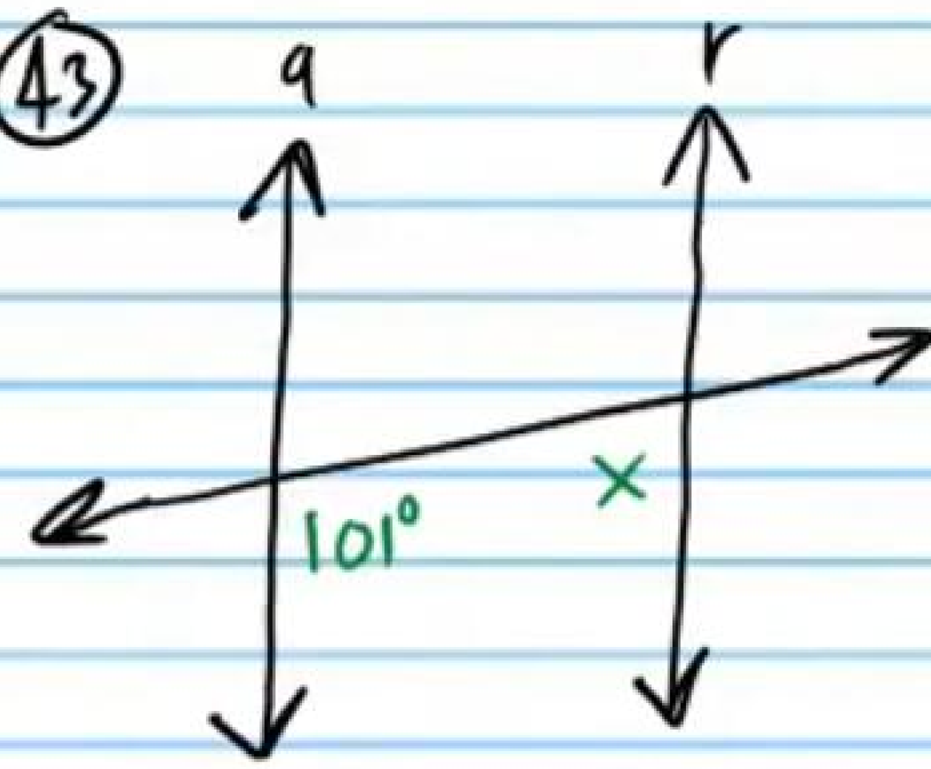
$x = 87°$ $\boxed{\text{AEACT}}$

(42)



$\boxed{y = 140°}$ $\boxed{\text{CACP}}$

(43)



$101 + x = 180$ $\boxed{\text{CIACT}}$

$\boxed{x = 79°}$